# Linear confinement and stress-energy tensor around static quark and anti-quark pair
## – Lattice simulation with Yang-Mills gradient flow –

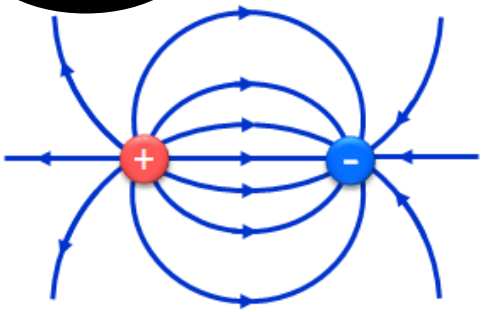
( RY *et al.*, arXiv : 1803.05656 [hep-lat]. )

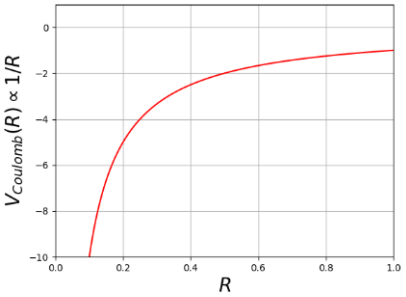
**Ryosuke Yanagihara (Osaka University)**

for FlowQCD collaboration :
Takumi Iritani, Masakiyo Kitazawa,
Masayuki Asakawa, Tetsuo Hatsuda

# QED vs. QCD

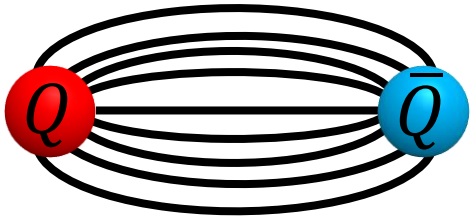
## QED

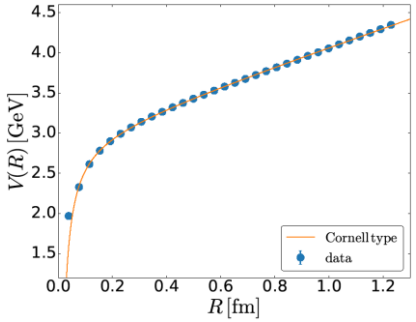



- ✓ Electric field spreads all over the space
- ✓ Coulomb potential

## QCD

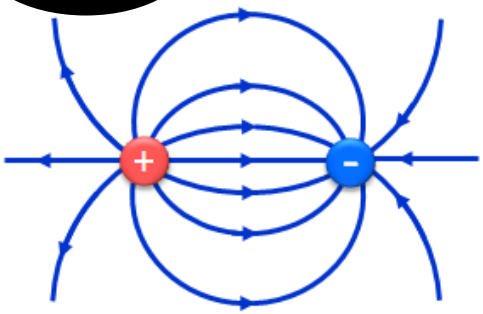


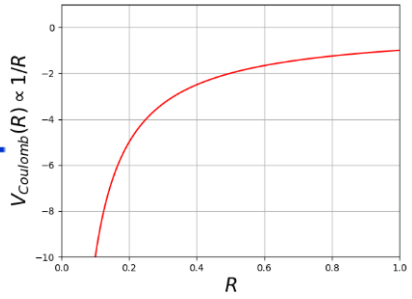




- ✓ Flux tube, squeezed one-dimensionally
- ✓ Confinement potential

# QED vs. QCD

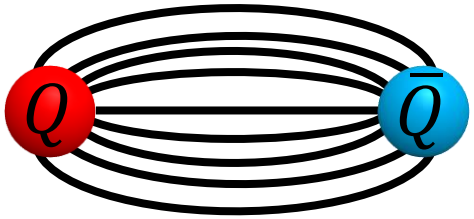
## QED

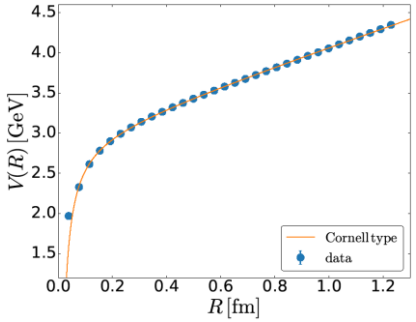



- ✓ Electric field spreads all over the space
- ✓ Coulomb potential

## QCD

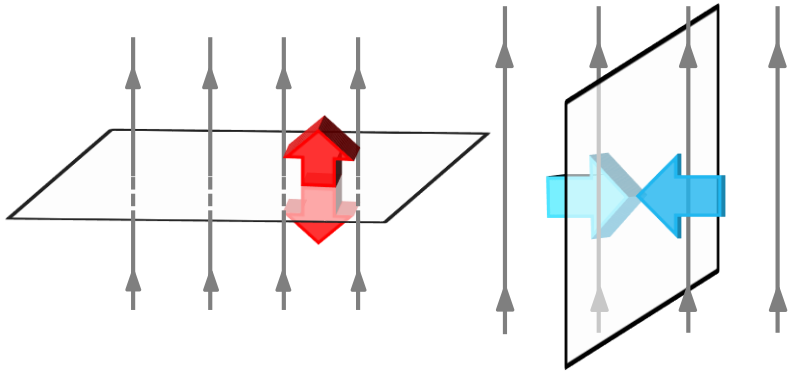



- ✓ Flux tube, squeezed one-dimensionally
- ✓ Confinement potential

## Local interaction

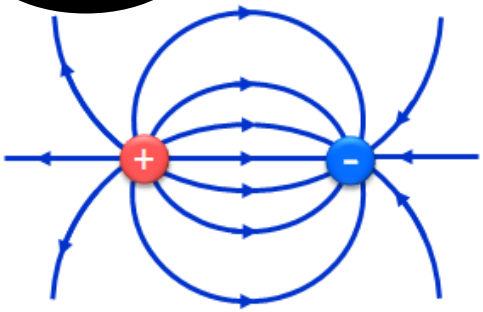
### Maxwell stress

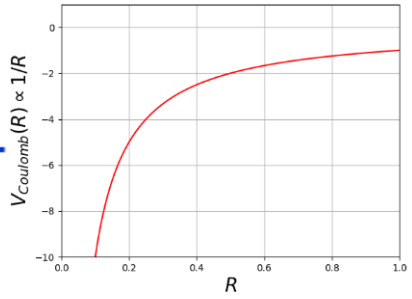


- ✓ Perpendicular plane : pulling
- ✓ Parallel plane : pushing

# QED vs. QCD

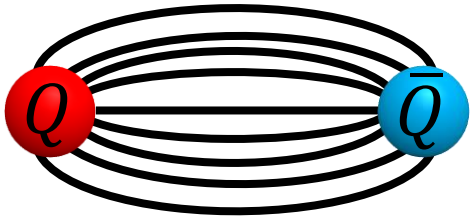
## QED

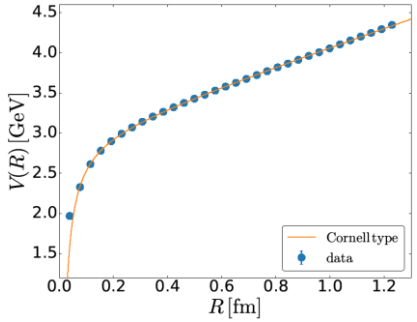



- ✓ Electric field spreads all over the space
- ✓ Coulomb potential

## QCD

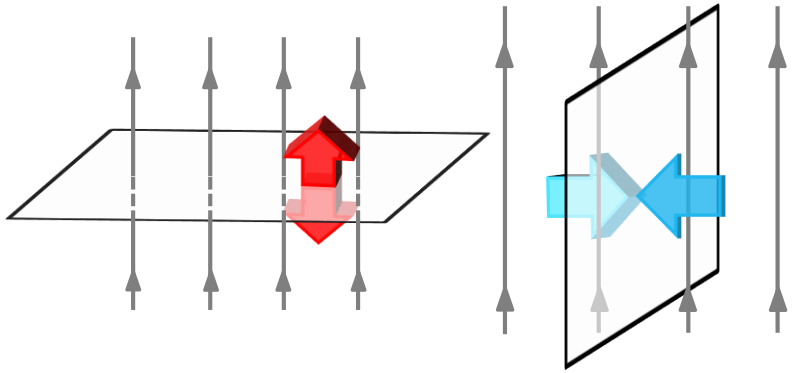



- ✓ Flux tube, squeezed one-dimensionally
- ✓ Confinement potential

## Local interaction

### Maxwell stress



- ✓ Perpendicular plane : pulling
- ✓ Parallel plane : pushing



# Energy Momentum Tensor (EMT)

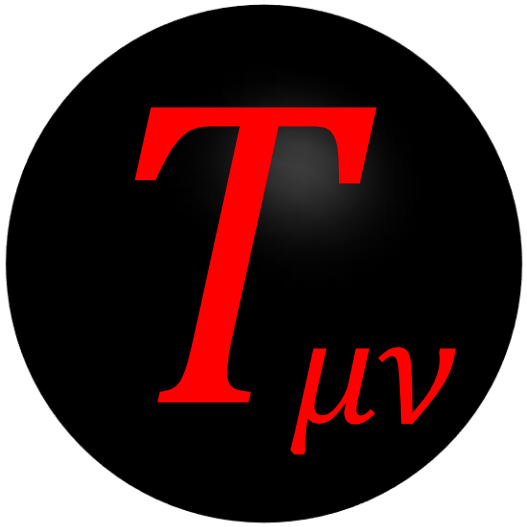
goal

Physics around $Q\bar{Q}$ in terms of energy and stress

# Energy Momentum Tensor (EMT)

goal

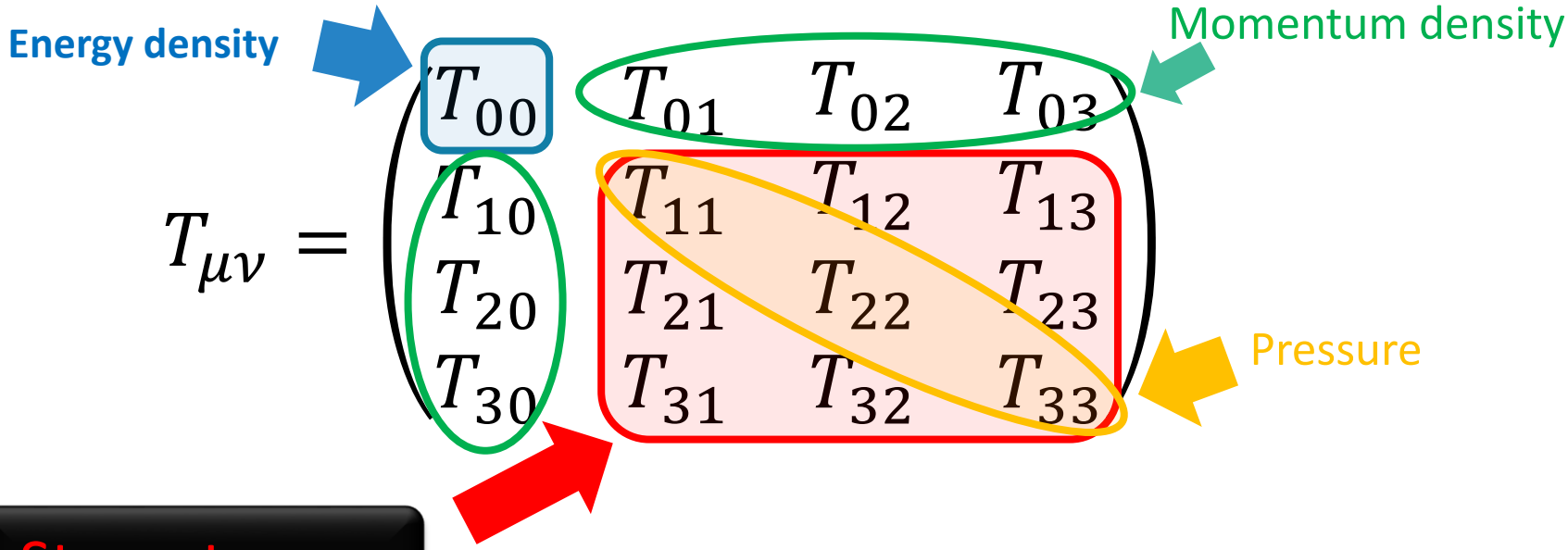
Physics around $Q\bar{Q}$ in terms of energy and stress



# Energy Momentum Tensor (EMT)

goal

**Physics around $Q\bar{Q}$ in terms of energy and stress**




✓ **Stress is force per unit area**

$$f_i = \sigma_{ij} n_j \ ; \ \sigma_{ij} = -T_{ij}$$

Landau and Lifshitz

# Energy Momentum Tensor (EMT)

goal

**Physics around $Q\bar{Q}$ in terms of energy and stress**

Energy density

Momentum density

$$T_{\mu\nu} = \begin{pmatrix} T_{00} & T_{01} & T_{02} & T_{03} \\ T_{10} & T_{11} & T_{12} & T_{13} \\ T_{20} & T_{21} & T_{22} & T_{23} \\ T_{30} & T_{31} & T_{32} & T_{33} \end{pmatrix}$$

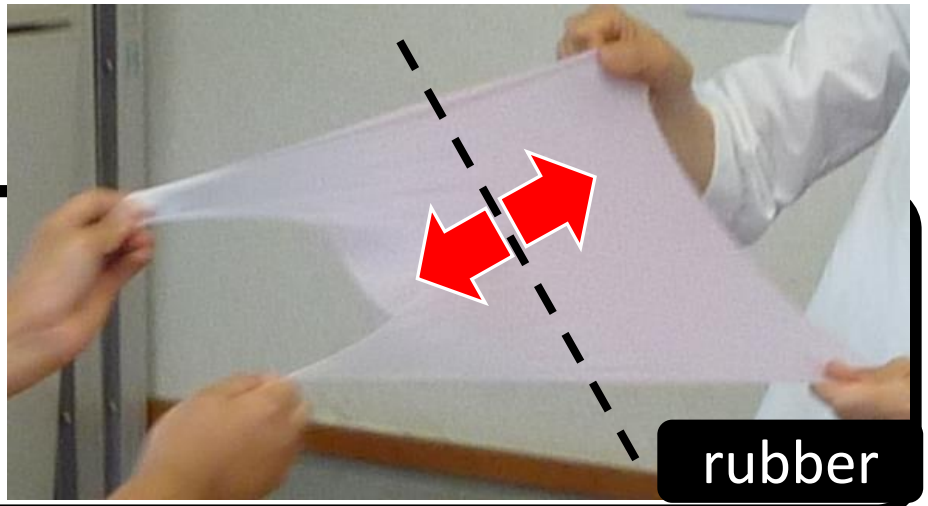
Pressure

Stress tensor

- ✓ **Stress is force per unit area**

$$f_i = \sigma_{ij} n_j \ ; \ \sigma_{ij} = -T_{ij}$$

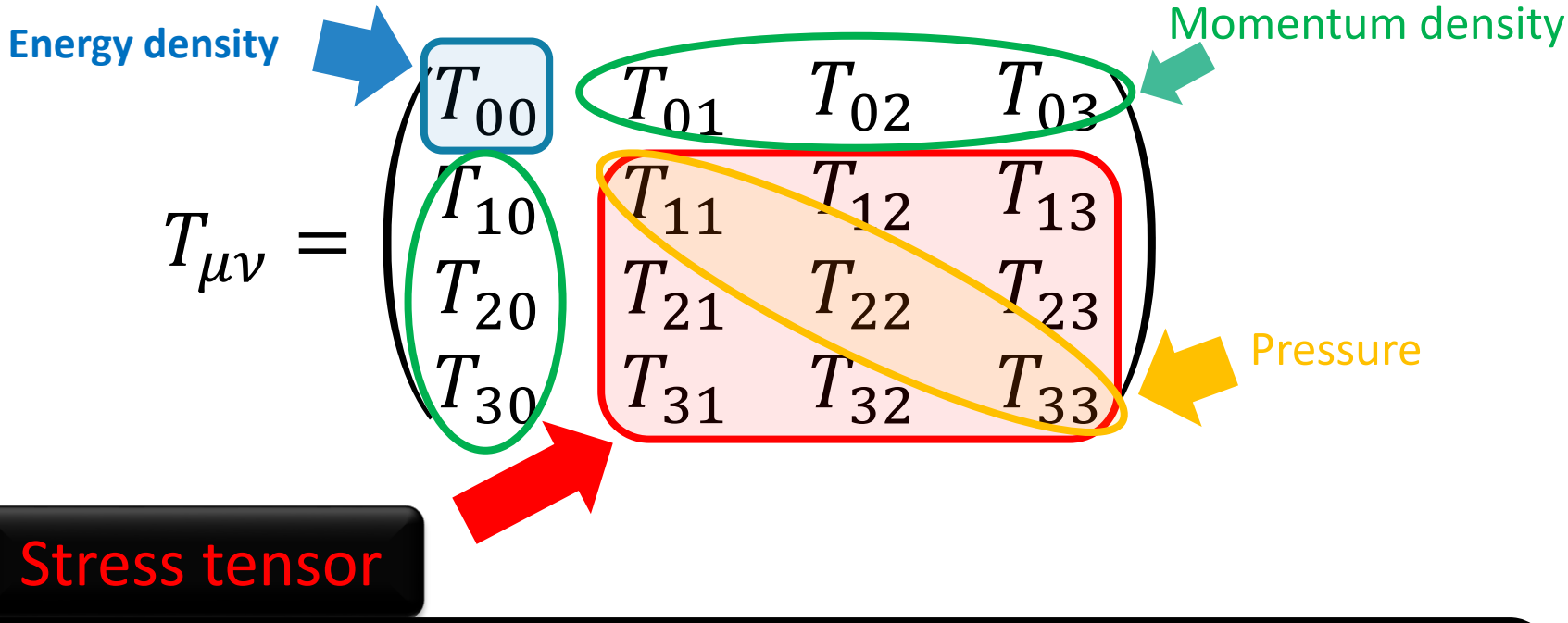
Landau and Lifshitz



# Energy Momentum Tensor (EMT)

goal

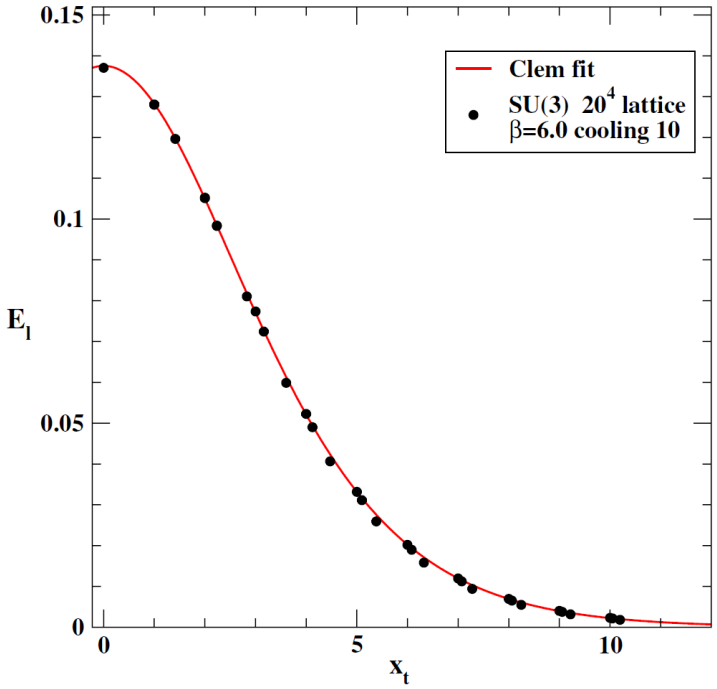
**Physics around $Q\bar{Q}$ in terms of energy and stress**




- ✓ **Gauge invariant !**
- ✓ Determine **absolute values** of all components

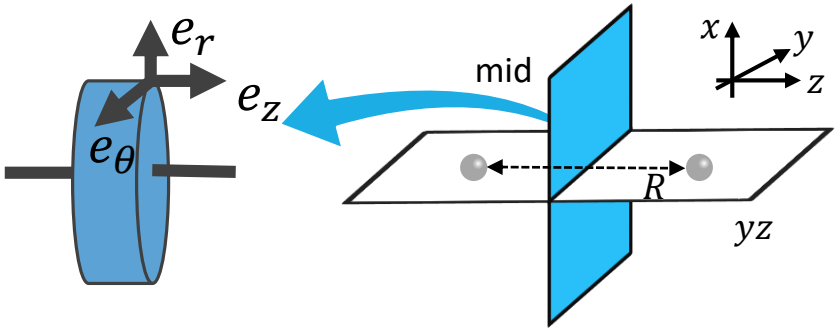
→ Local Interaction

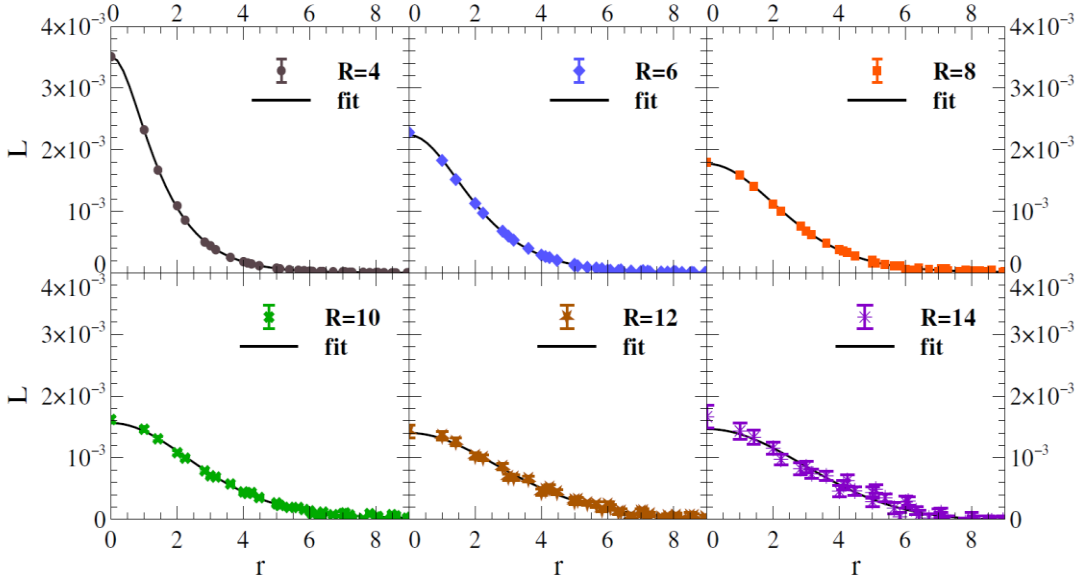
# A lot of Previous Studies




Color electric field

Cea *et al.*, PRD**88** (2012) 054504.

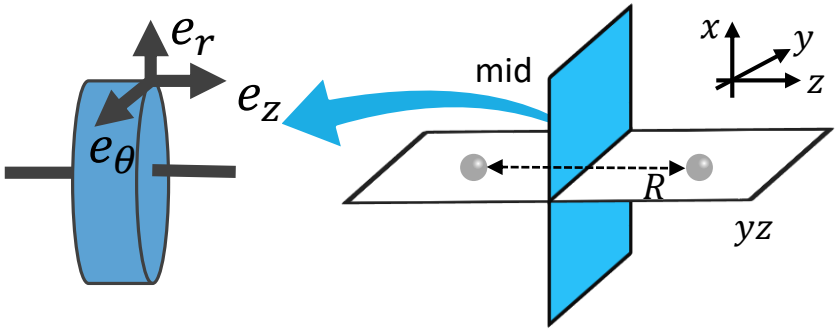


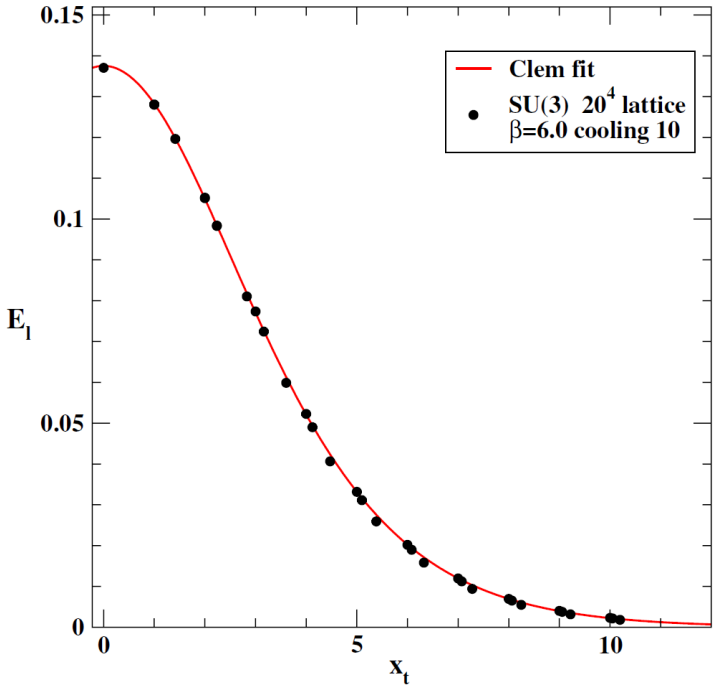



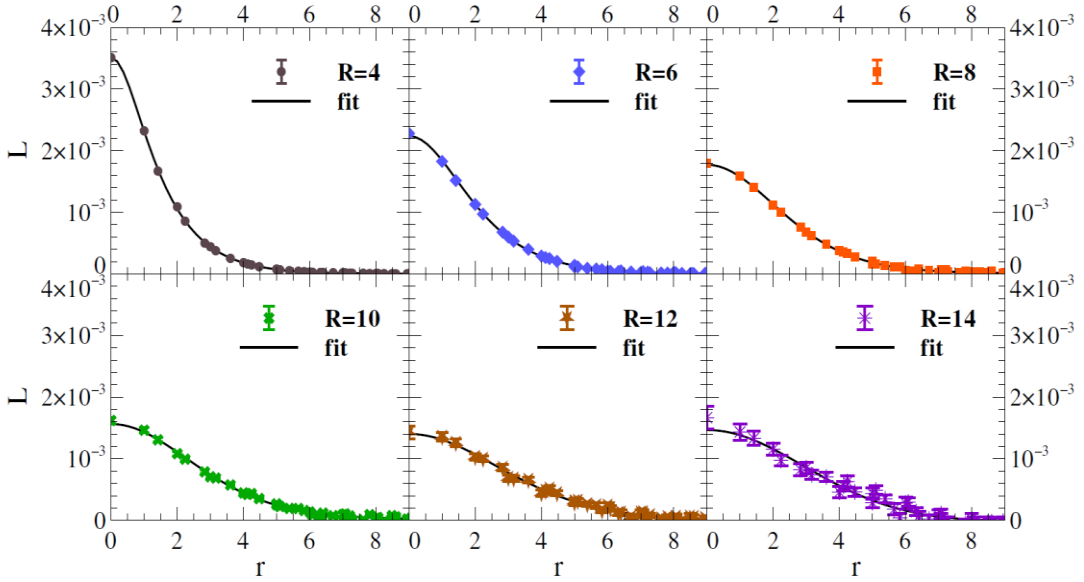

Action density

Cardoso *et al.*, PRD**86** (2013) 054501.

# A lot of Previous Studies




Color electric field

Cea *et al.*, PRD**88** (2012) 054504.

Action density

Cardoso *et al.*, PRD**86** (2013) 054501.

More direct physical quantity : Stress tensor !!

# Measurement of the stress on the Lattice

To Do

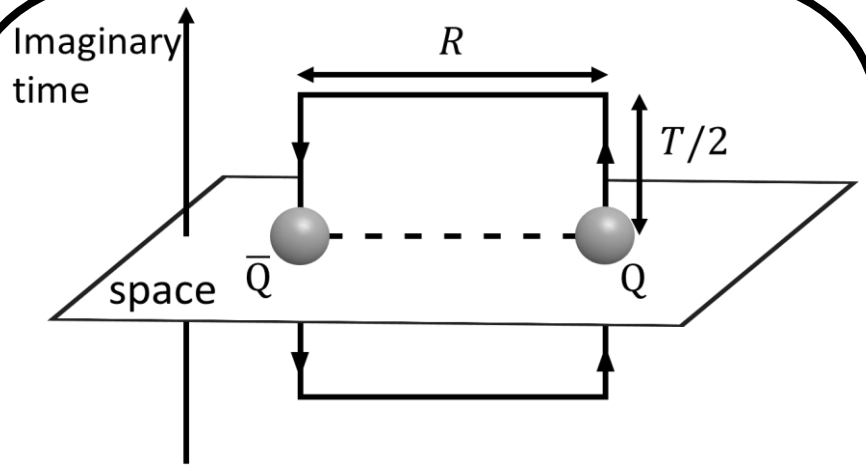
①prepare $Q\bar{Q}$ on the lattice and ②measure EMT around $Q\bar{Q}$

# Measurement of the stress on the Lattice

To Do

①prepare $Q\bar{Q}$ on the lattice and ②measure EMT around $Q\bar{Q}$

## Wilson Loop

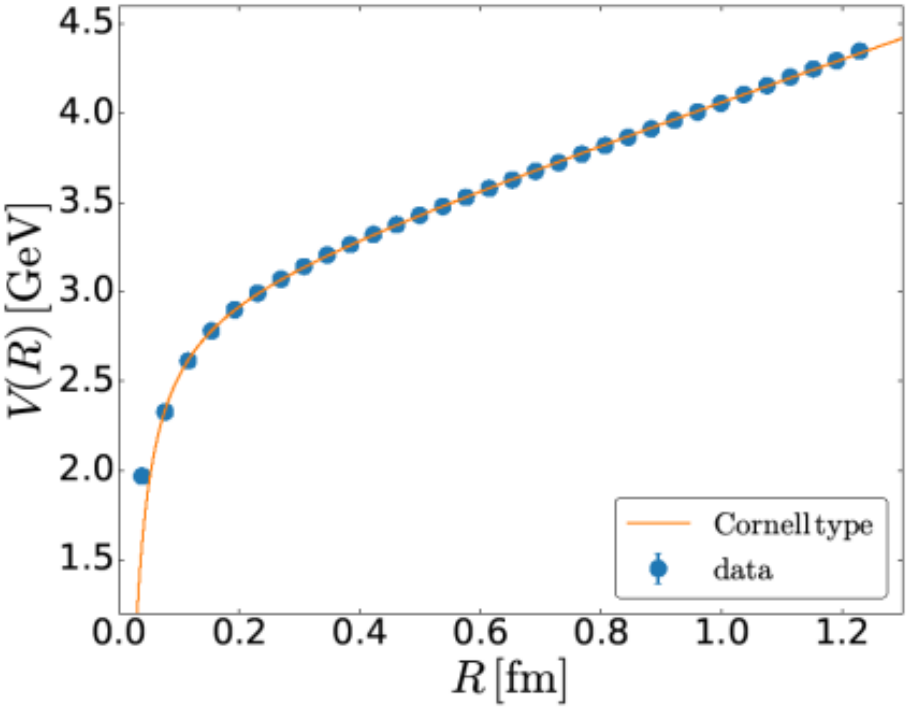



$$\langle W(R,T)\rangle = C_0\exp[-V(R)T] + C_1\exp[-V_1(R)T] + \cdots$$

$$V(R) = -\lim_{T\to\infty}\frac{1}{T}\log\langle W(R,T)\rangle$$

Ground state potential

Confinement potential




- ✓ quenched SU(3) Yang-Mills
- ✓ $\beta = 6.600$ ($a = 0.038$ fm)

# Measurement of the stress on the Lattice

To Do

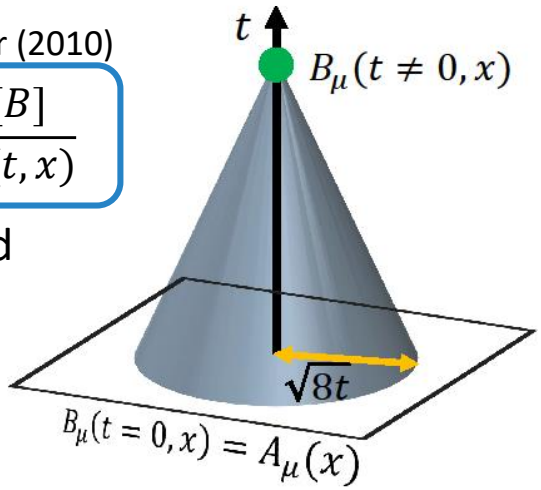
①prepare $Q\bar{Q}$ on the lattice and ②measure EMT around $Q\bar{Q}$

## Gradient flow

Flow eq. Lüscher (2010)

$$\frac{\partial B_\mu(t,x)}{\partial t} = -g_0^2 \frac{\delta S[B]}{\delta B_\mu(t,x)}$$

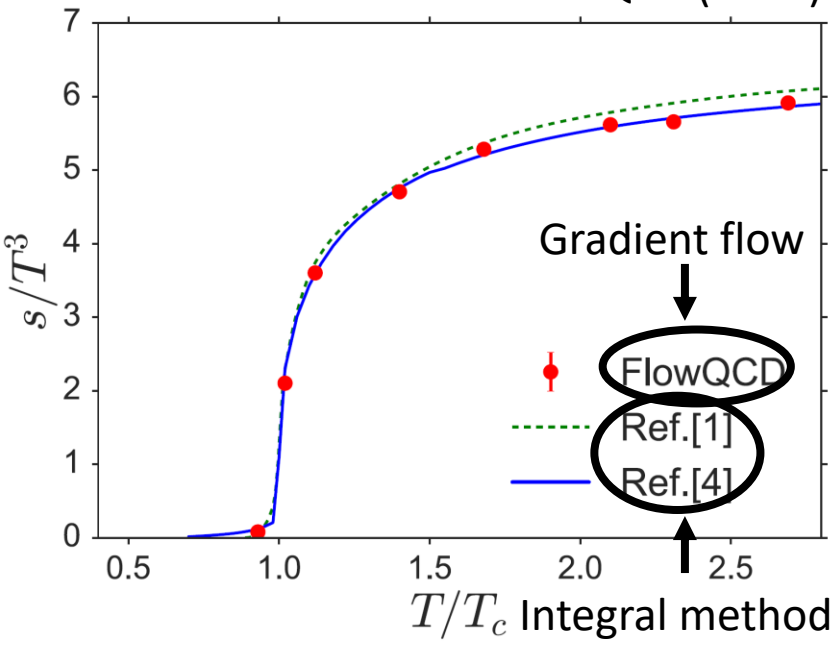
$B_\mu$ : smeared field




EMT defined via gradient flow Suzuki (2013)

$$T_{\mu\nu}(t,x) = \frac{1}{\alpha_U(t)} U_{\mu\nu}(t,x) + \frac{\delta_{\mu\nu}}{4\alpha_E(t)} [E(t,x) - \langle E(t,x)\rangle] + O(t)$$

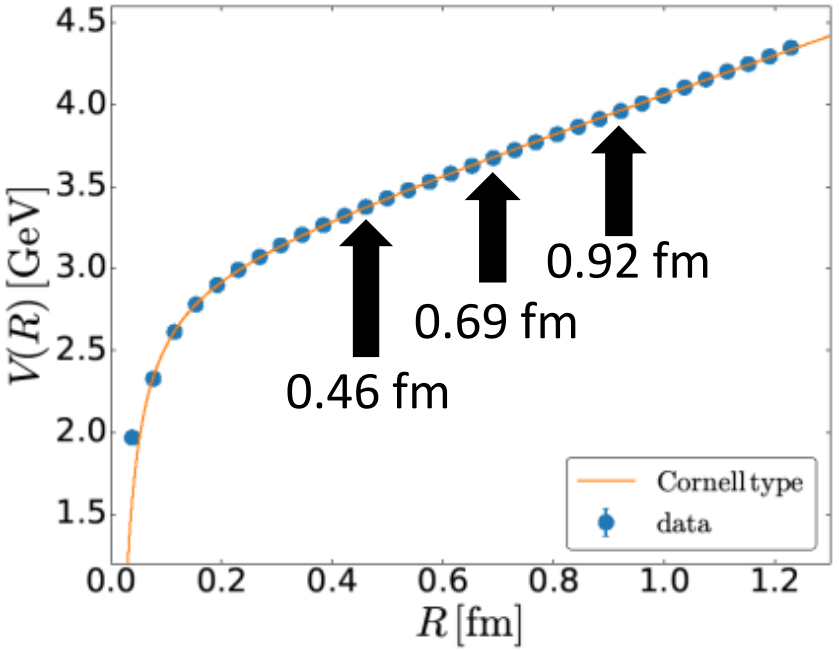
FlowQCD (2016)




Entropy density vs. temperature

# Setup

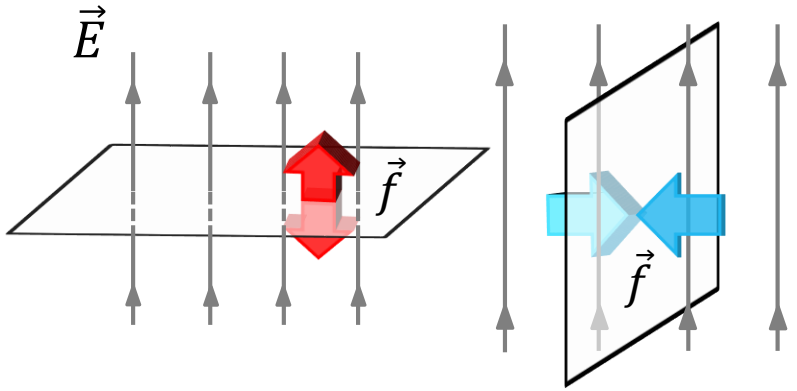
- ✓ Quenched SU(3) Yang-Mills gauge theory
- ✓ Wilson gauge action
- ✓ Clover operator
- ✓ Continuum limit
- ✓ APE smearing for spatial links
- ✓ Multihit improvement in temporal links
- ✓ Simulation using BlueGene/Q @ KEK

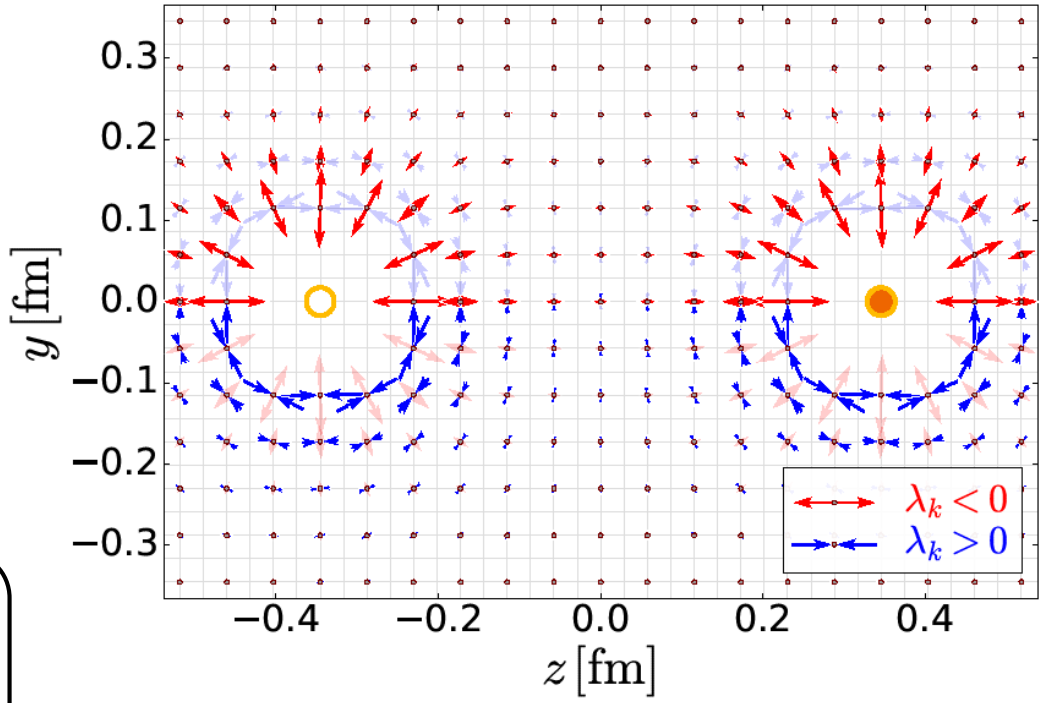



| $\beta$ | Lattice spacing | Lattice size | # of statistics |
|---|---|---|---|
| 6.304 | 0.057 fm | $48^4$ | 140 |
| 6.465 | 0.046 fm | $48^4$ | 440 |
| 6.513 | 0.043 fm | $48^4$ | 600 |
| 6.600 | 0.038 fm | $48^4$ | 1500 |
| 6.819 | 0.029 fm | $64^4$ | 1000 |

# Stress distribution in Maxwell theory

$$T_{ij} = \epsilon_0 \left( E_i E_j - \frac{\delta_{ij}}{2} E^2 \right) + \frac{1}{\mu_0} \left( B_i B_j - \frac{\delta_{ij}}{2} B^2 \right)$$




- ✓ Perpendicular plane: $\lambda_k < 0$
- ✓ Parallel plane: $\lambda_k > 0$

✓ Stress tensor

$$T_{ij} n_j^{(k)} = \lambda_k n_i^{(k)}$$

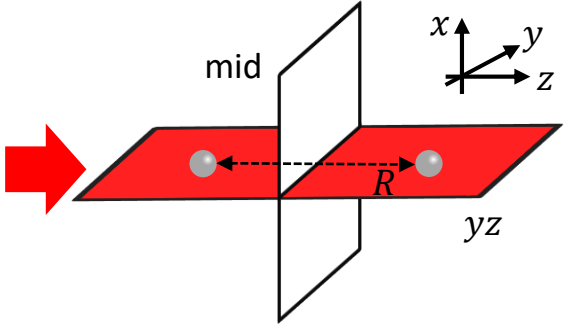
$$(i, j = 1,2,3 \; ; \; k = 1,2,3)$$

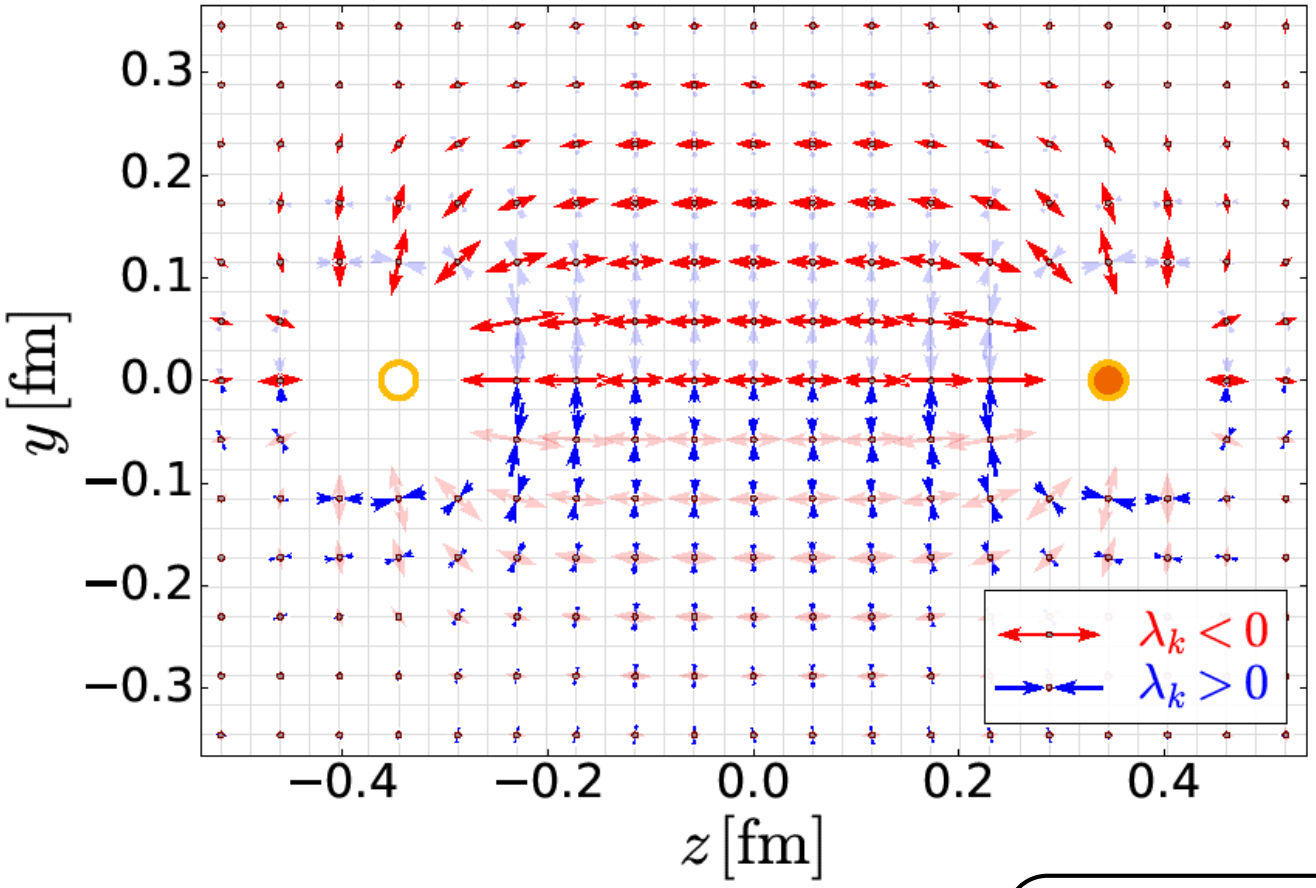
$$f_i = \sigma_{ij} n_j \; ; \; \sigma_{ij} = -T_{ij}$$

Landau and Lifshitz

Note that the length of the arrow is equal to $\sqrt{\lambda_k}$

# Stress distribution in SU(3) YM theory

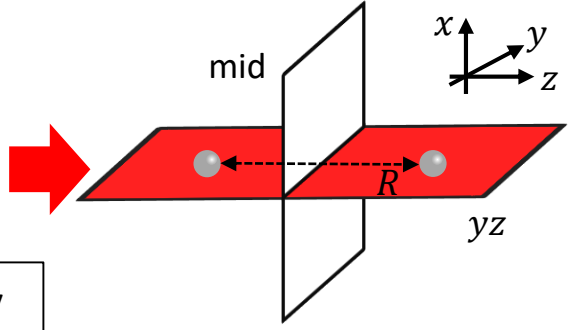


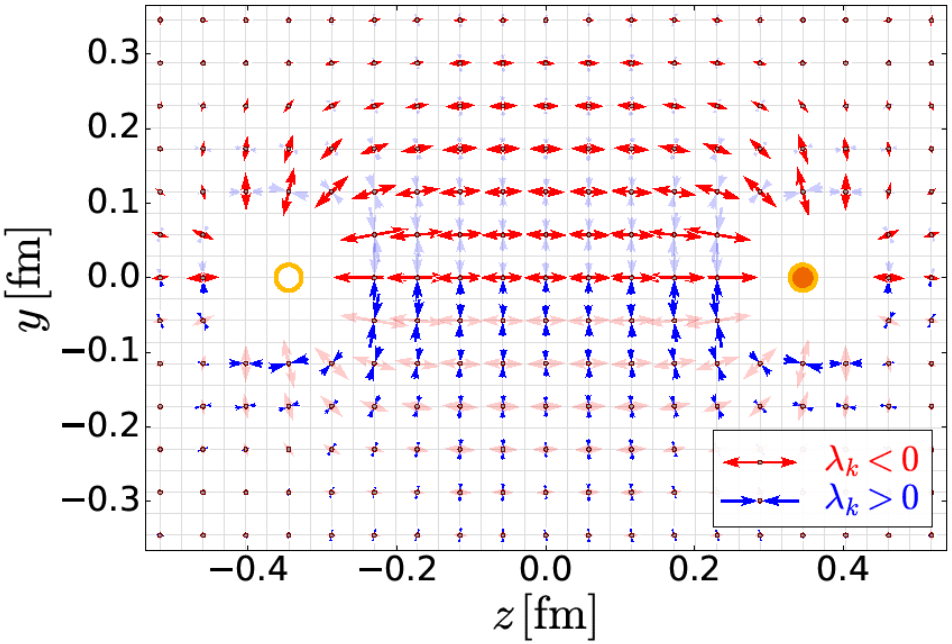

Note that the length of the arrow is equal to $\sqrt{\lambda_k}$

- ✓ $a = 0.029$ fm(no continuum limit)
- ✓ $t/a^2 = 2.0$ (no $t \to 0$ limit)
- ✓ $R = 0.69$ fm

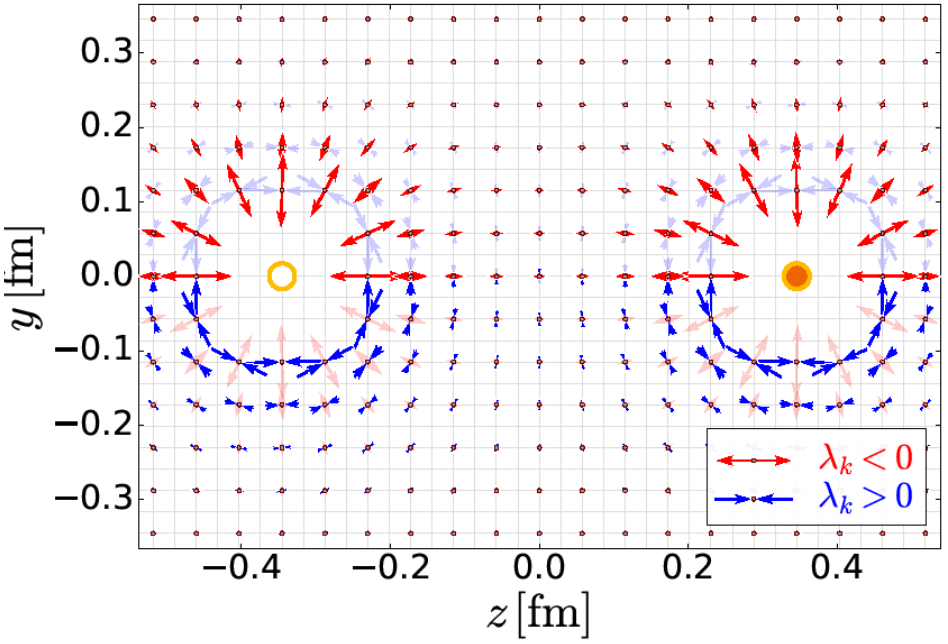
# SU(3) YM theory vs. Maxwell theory




SU(3) YM theory

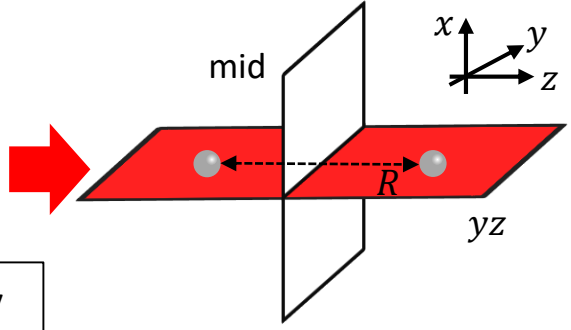



Maxwell theory

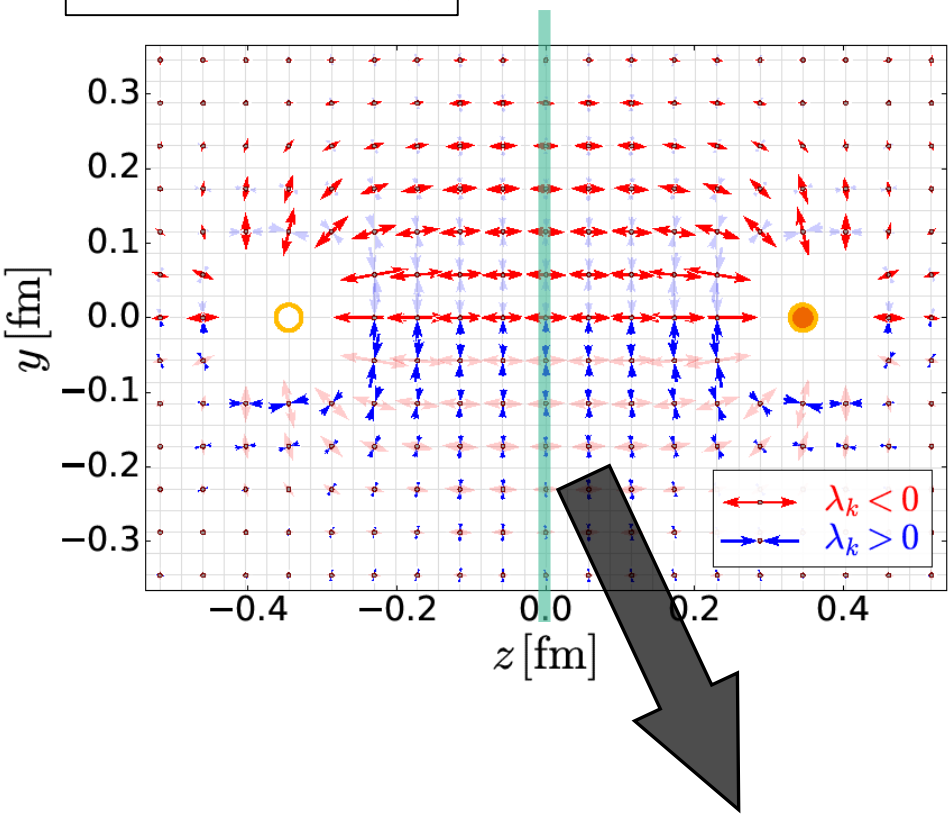



- ✓ Gauge invariant
- ✓ In terms of local interaction
- ✓ Propagation of force : squeezed vs. spreading

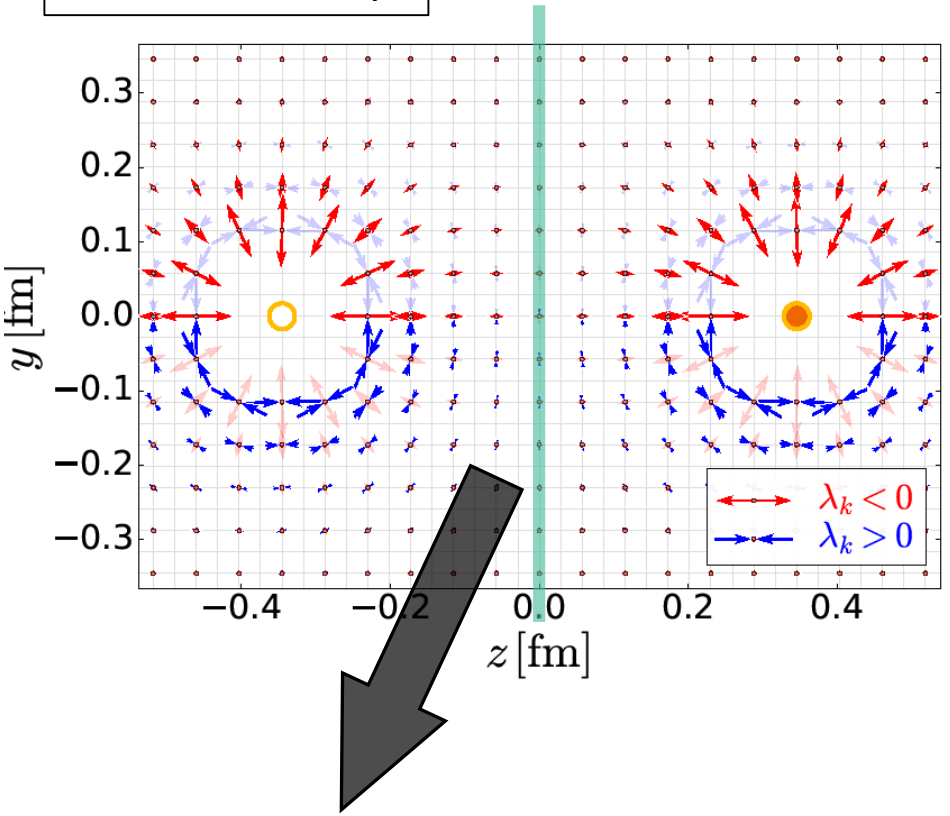
# SU(3) YM theory vs. Maxwell theory




SU(3) YM theory




Maxwell theory

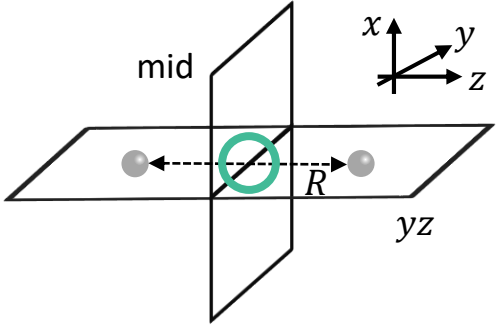


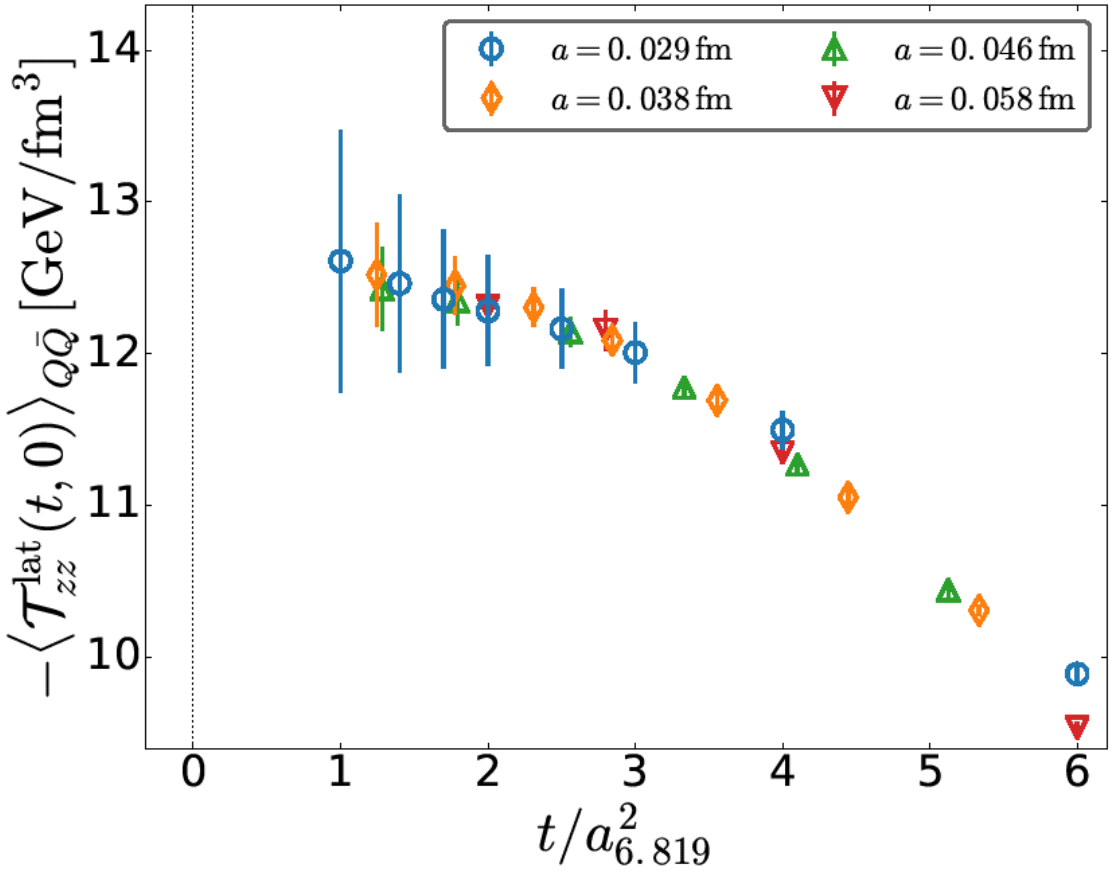

next

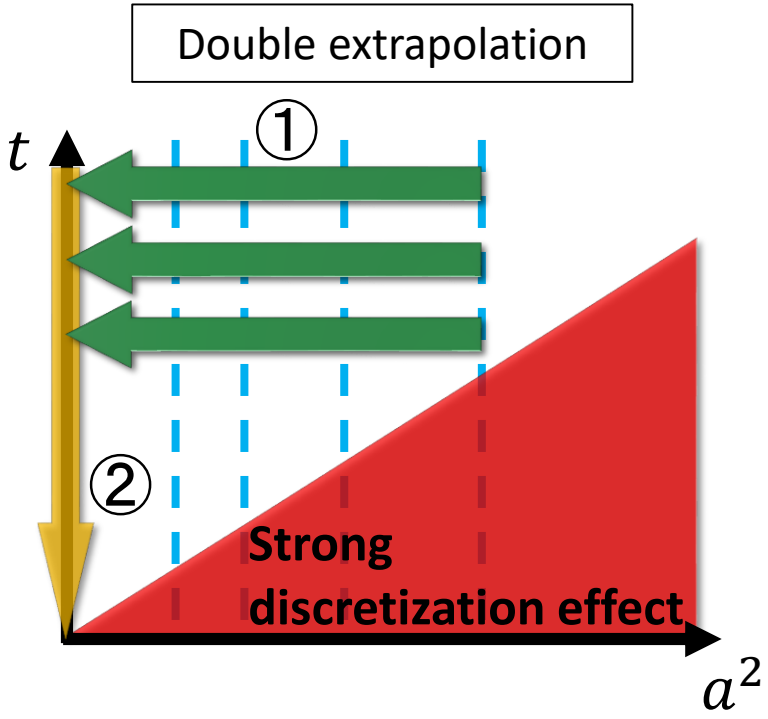
We focus on the mid-plane : double extrapolation

# Double extrapolation @ mid point ($R = 0.46$ fm)

FlowQCD (2016)



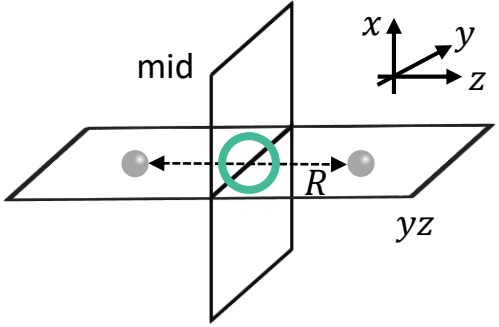



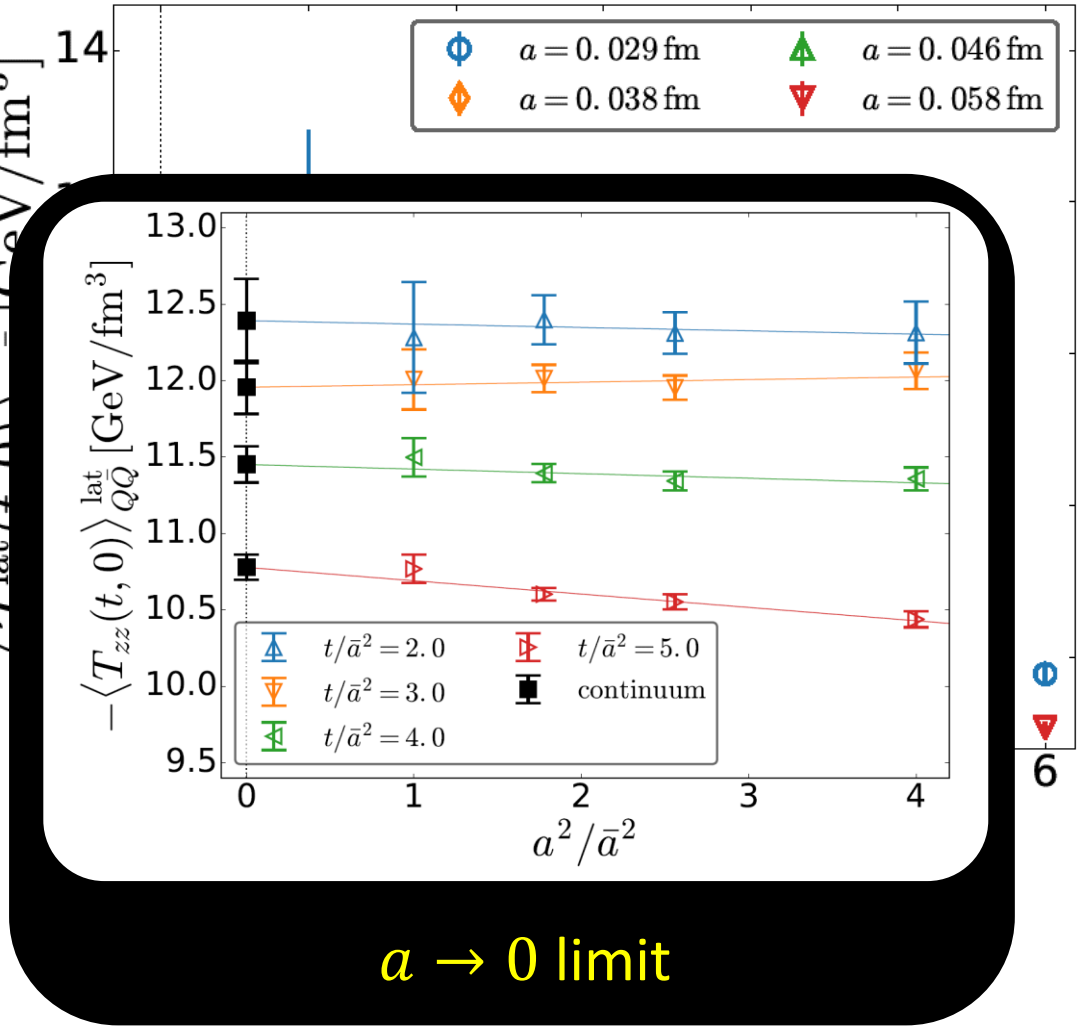

$$O_{\mathrm{lat}} = O_{\mathrm{cont}} + A_0 t + A_1(t)a^2 + \cdots$$

② ①

# Double extrapolation @ mid point ($R = 0.46$ fm)

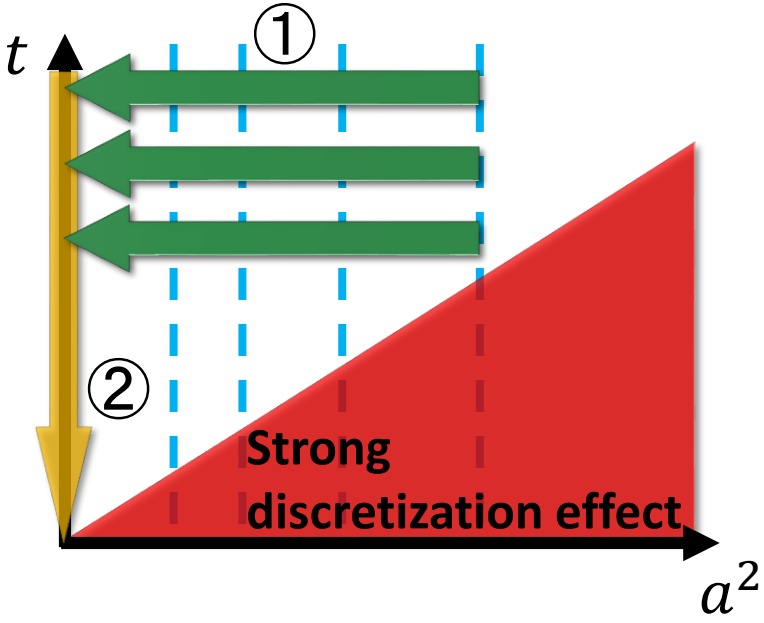
FlowQCD (2016)




$a \to 0$ limit

Double extrapolation

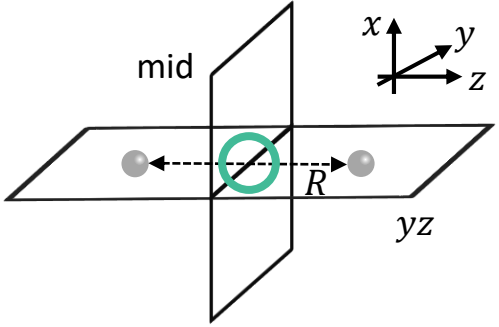


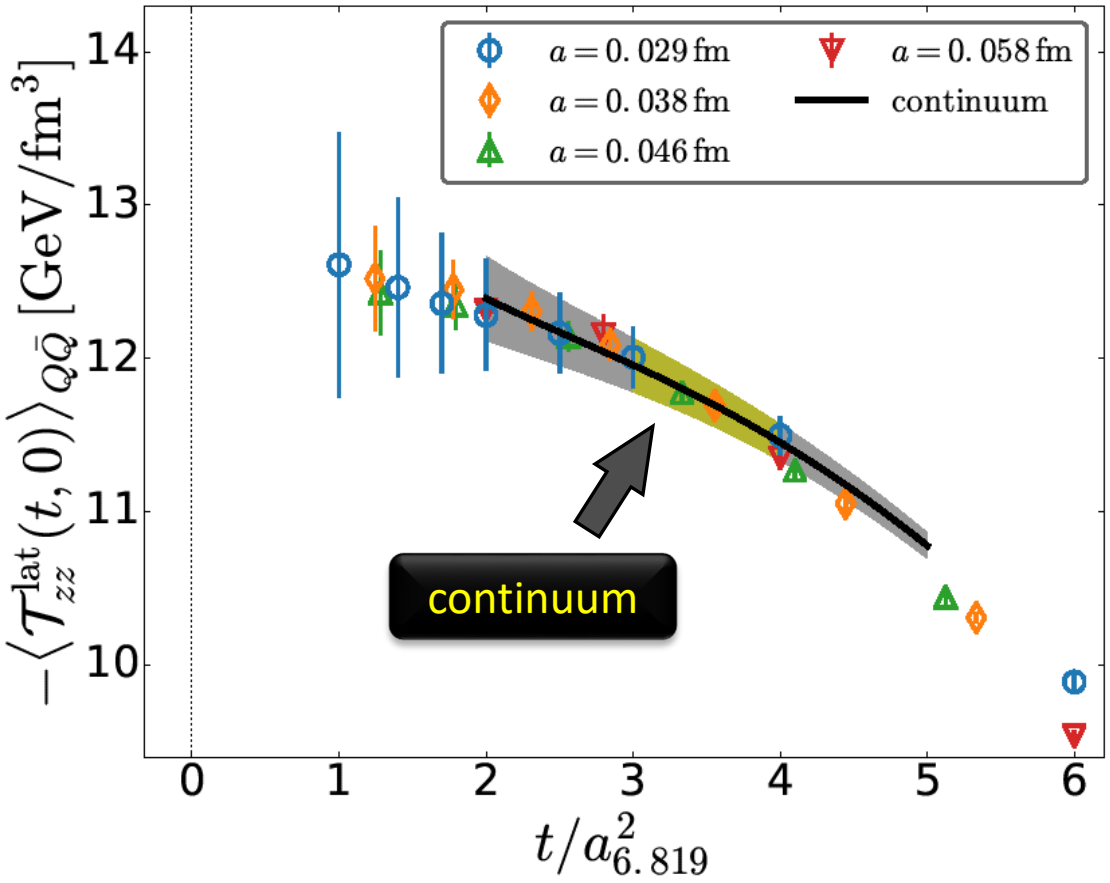

$$O_{\rm lat} = O_{\rm cont} + A_0 t + A_1(t) a^2 + \cdots$$

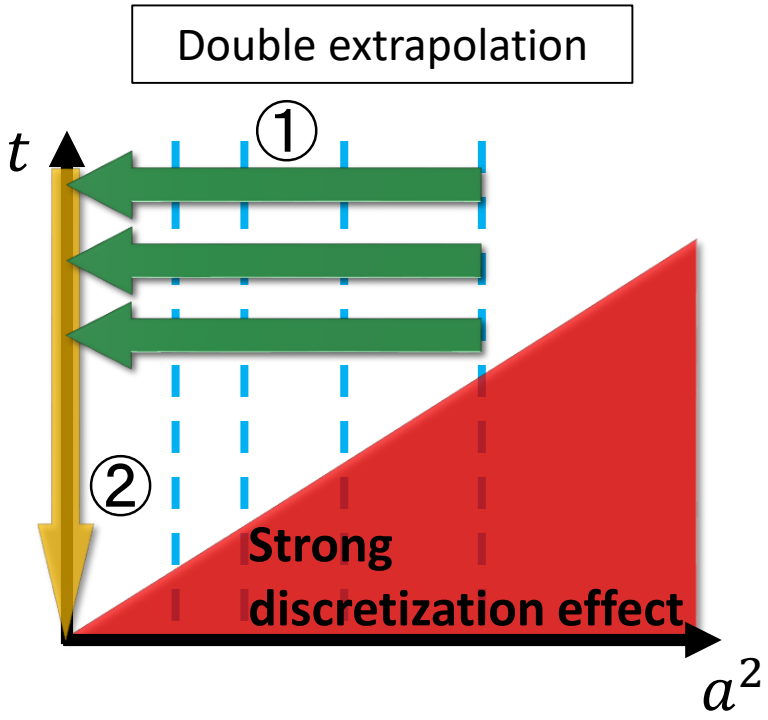
② ①

# Double extrapolation @ mid point ($R = 0.46$ fm)

FlowQCD (2016)



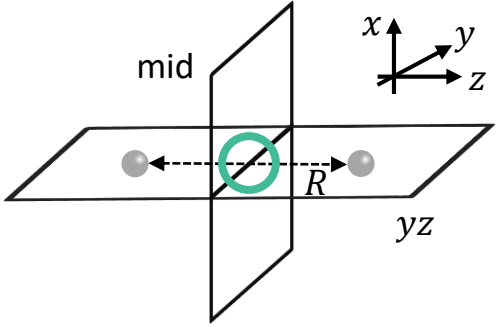



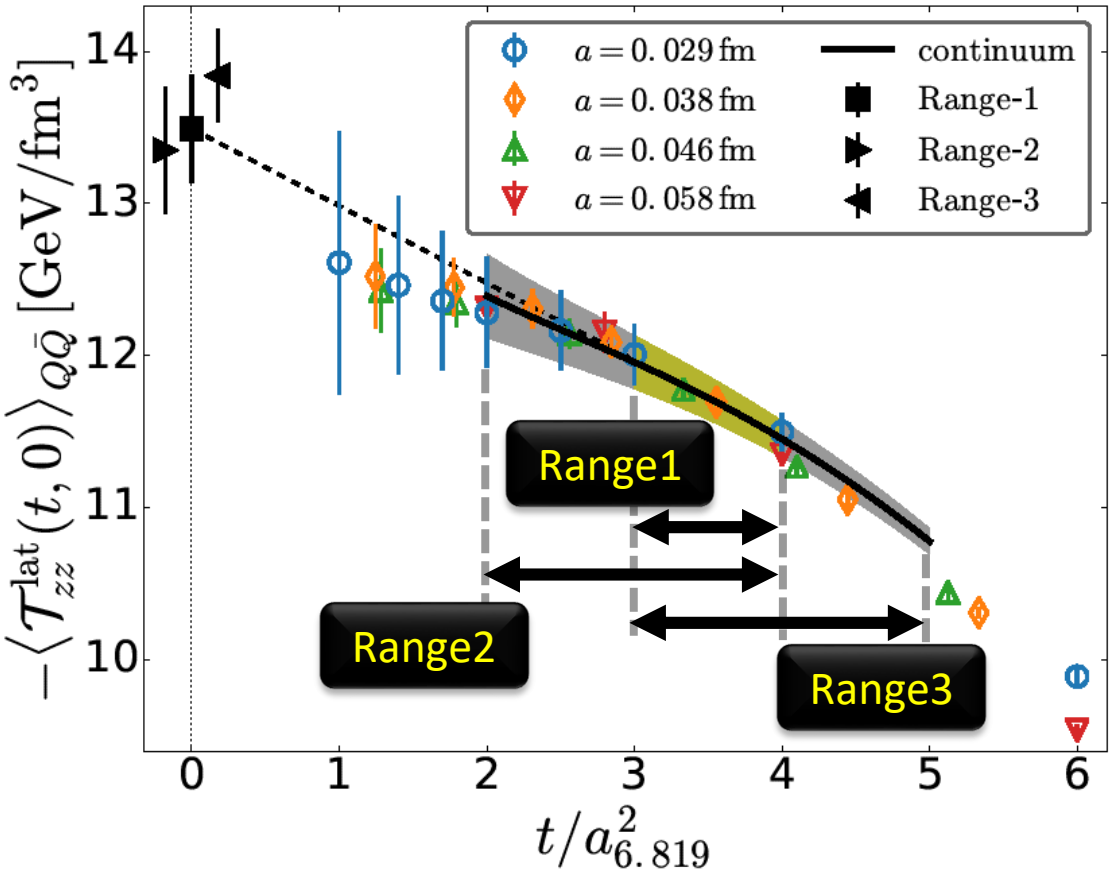

$$O_{\mathrm{lat}} = O_{\mathrm{cont}} + A_0 t + A_1(t) a^2 + \cdots$$

② ①

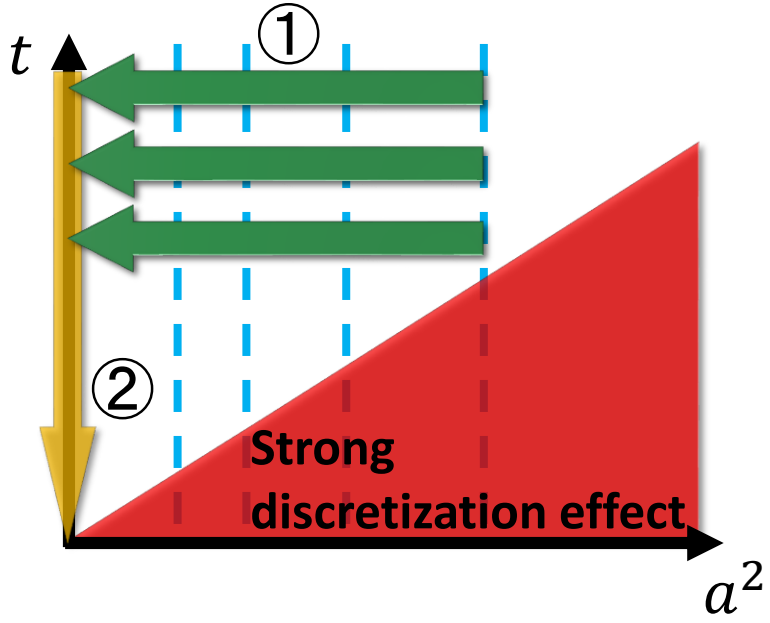
# Double extrapolation @ mid point ($R = 0.46$ fm)

FlowQCD (2016)



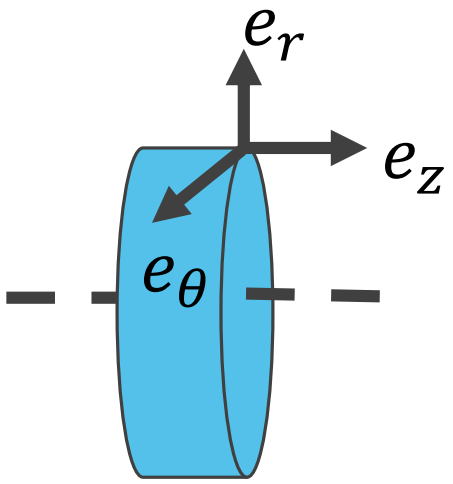

Double extrapolation

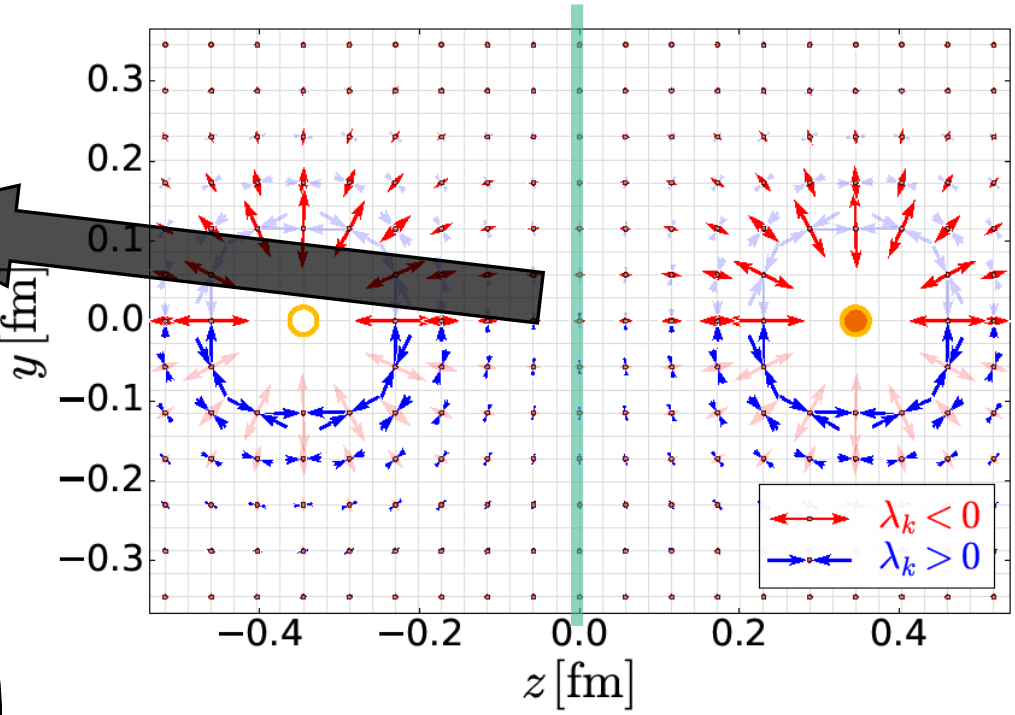



$$O_{\rm lat} = O_{\rm cont} + A_0 t + A_1(t)a^2 + \cdots$$

② ①

# Cylindrical coordinate



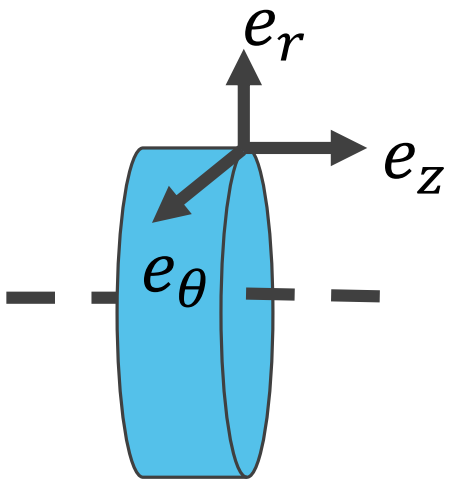



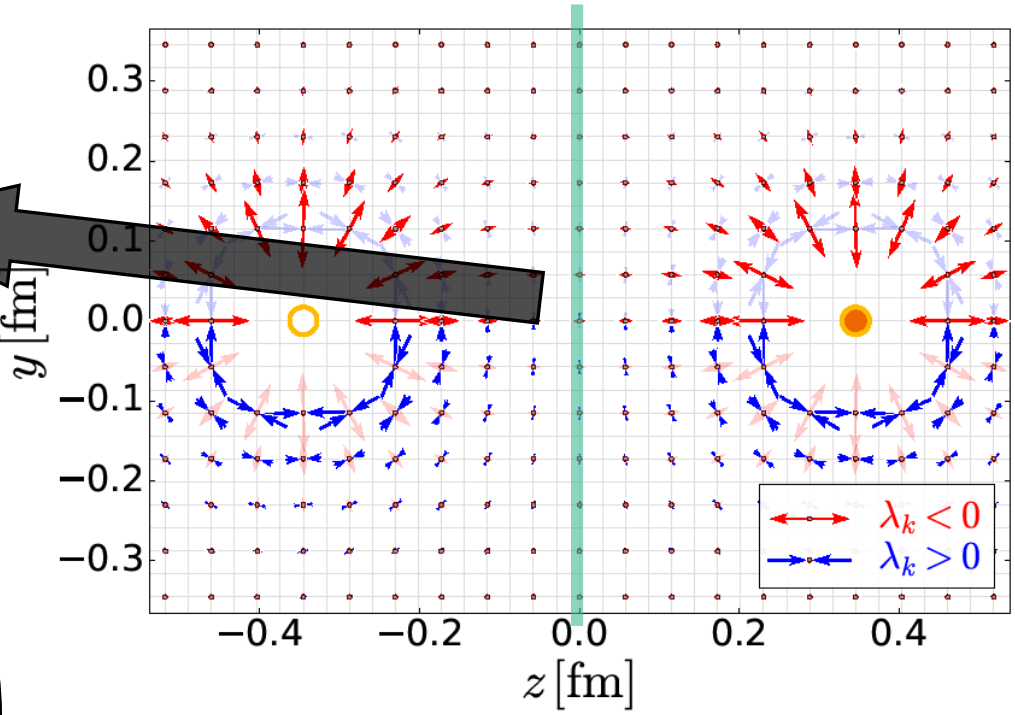

$$T_{\mu\nu} = \begin{pmatrix} T_{44} & & & O \\ & T_{zz} & & \\ & & T_{rr} & \\ O & & & T_{\theta\theta} \end{pmatrix}$$

**Diagonalization of EMT**

(Cylindrical symmetry and parity symmetry)

# Cylindrical coordinate




$$T_{\mu\nu} = \begin{pmatrix} T_{44} & & & O \\ & T_{zz} & & \\ & & T_{rr} & \\ O & & & T_{\theta\theta} \end{pmatrix}$$

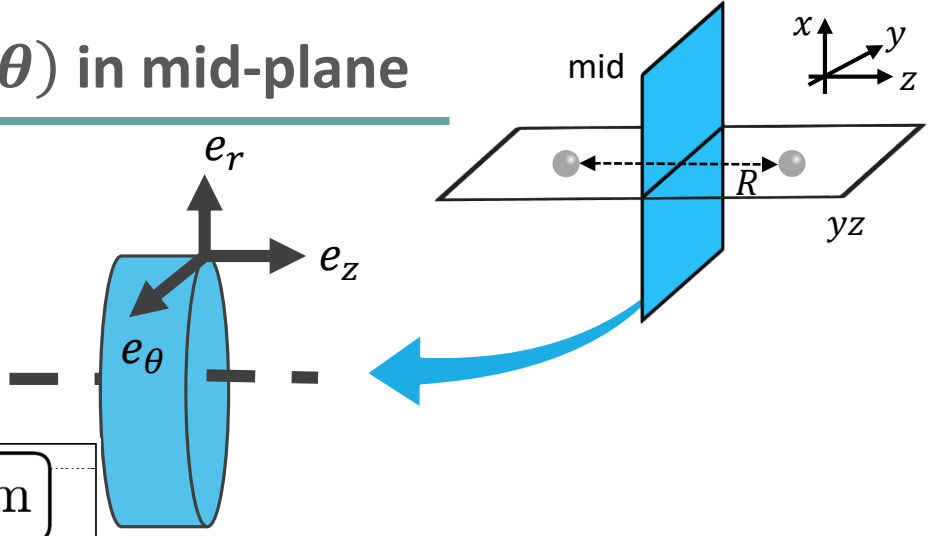
Diagonalization of EMT

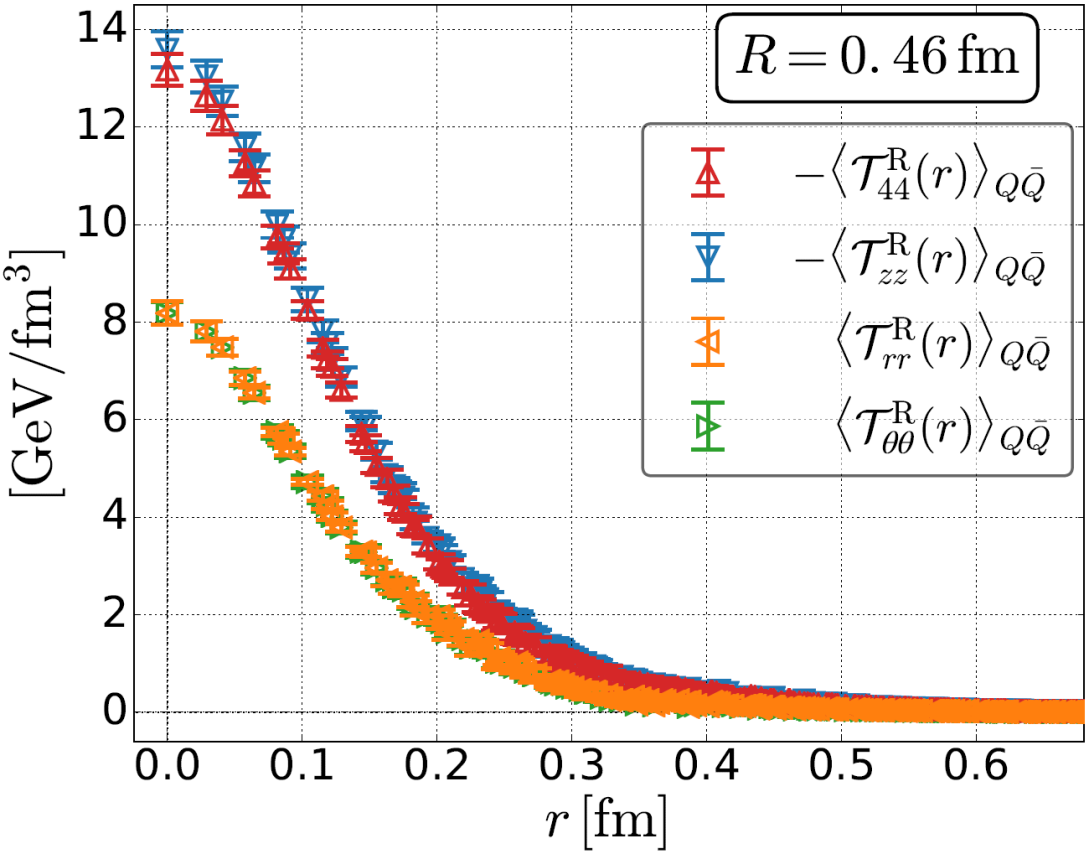
(Cylindrical symmetry and parity symmetry)

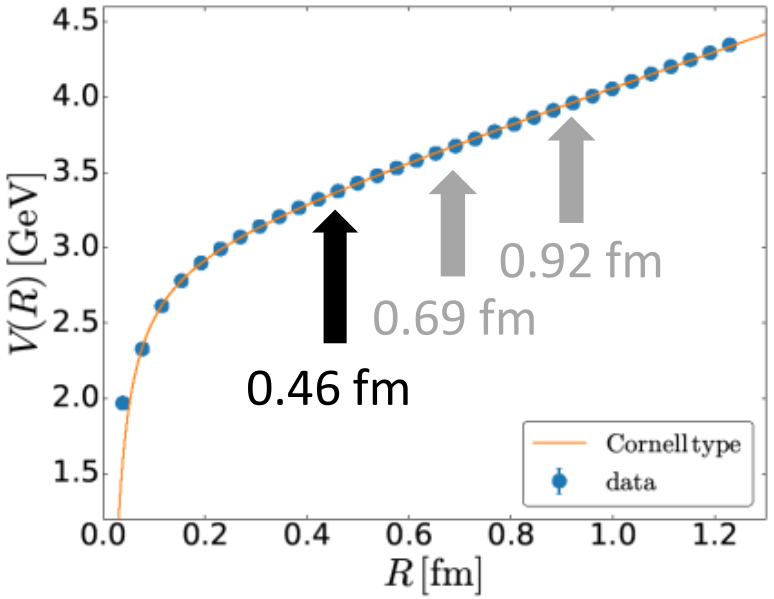
Degenerate in Maxwell theory !

$$-T_{44} = -T_{zz} = T_{rr} = T_{\theta\theta}$$

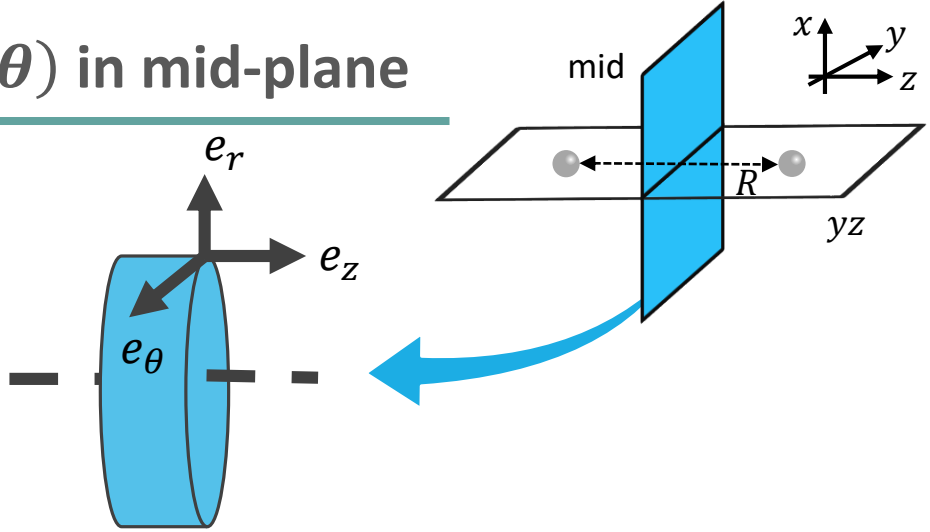
# Distribution of $\langle T_{cc} \rangle_{Q\bar{Q}}$ $(c = 4, z, r, \theta)$ in mid-plane

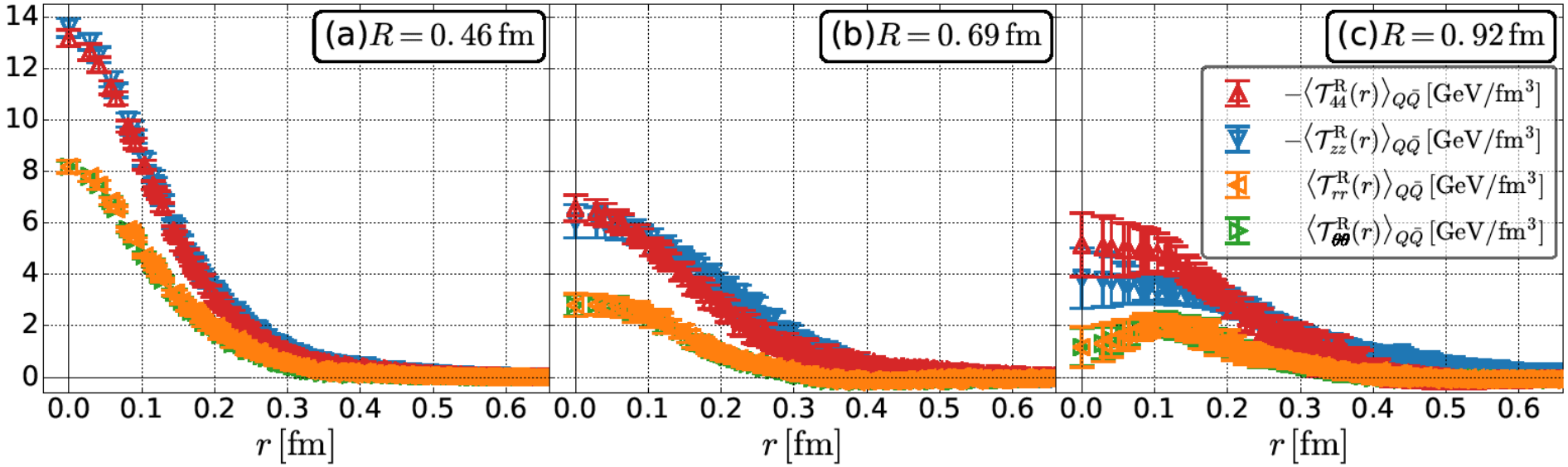


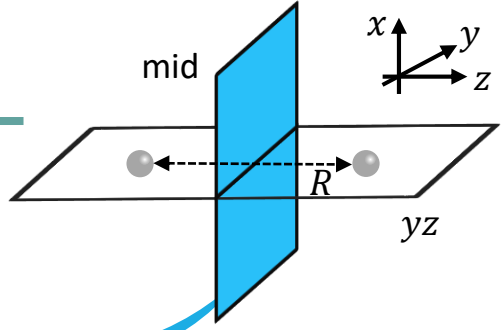



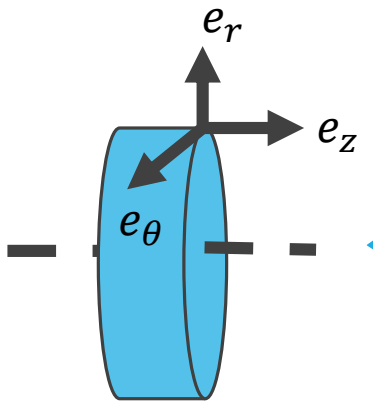
# Distribution of $\langle T_{cc} \rangle_{Q\bar{Q}}$ $(c = 4, z, r, \theta)$ in mid-plane



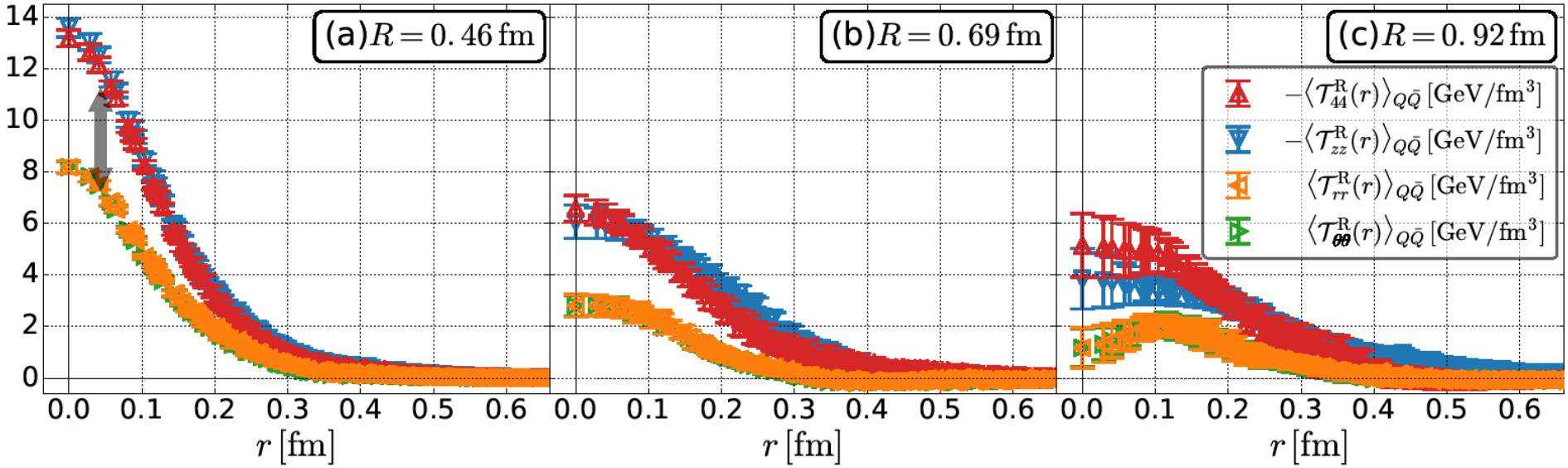
# Distribution of $\langle T_{cc} \rangle_{Q\bar{Q}}$ $(c = 4, z, r, \theta)$ in mid-plane



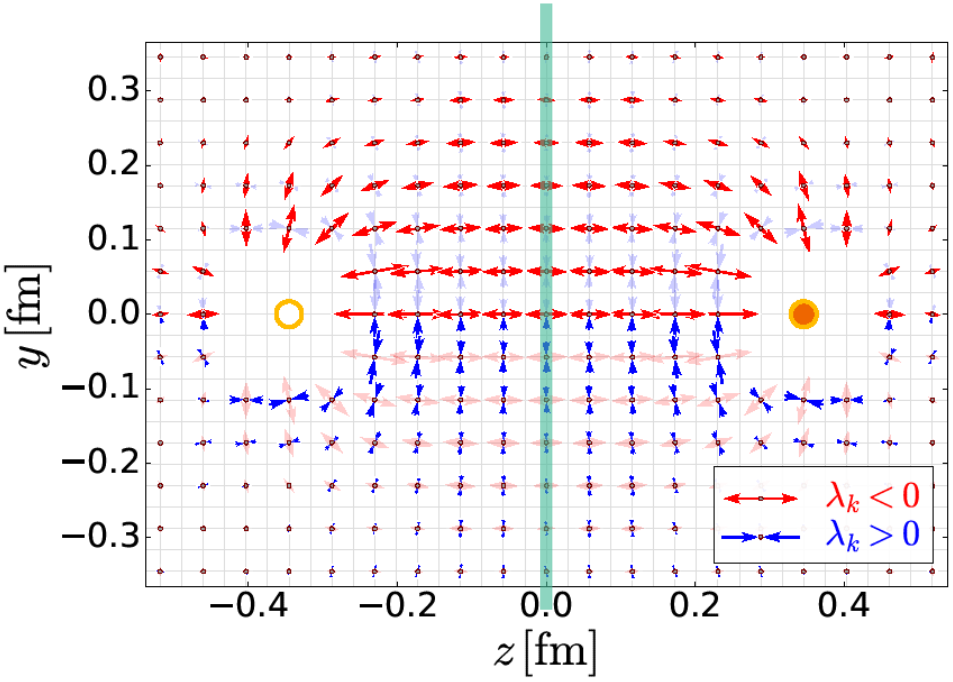

## Properties in non-Abelian theory

- ✓ $T_{44} \approx T_{zz}, T_{rr} \approx T_{\theta\theta}$ (degeneracy)
- ✓ $T_{44} \neq T_{rr}$ (separation)
- ✓ $\sum_\mu T_{\mu\mu} \neq 0$

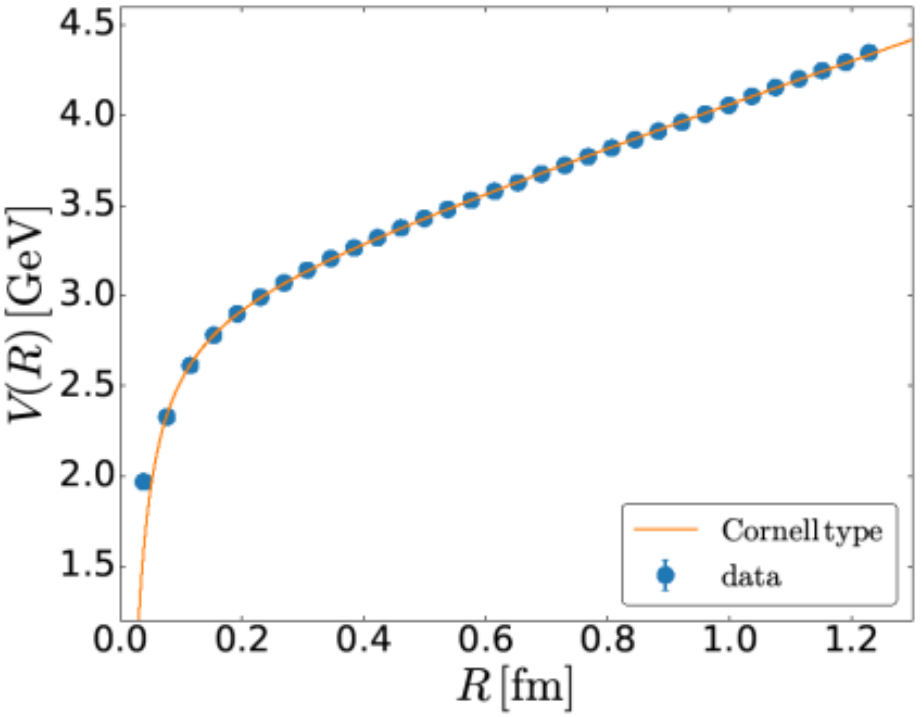


# EMT and confinement potential

From EMT




confinement potential



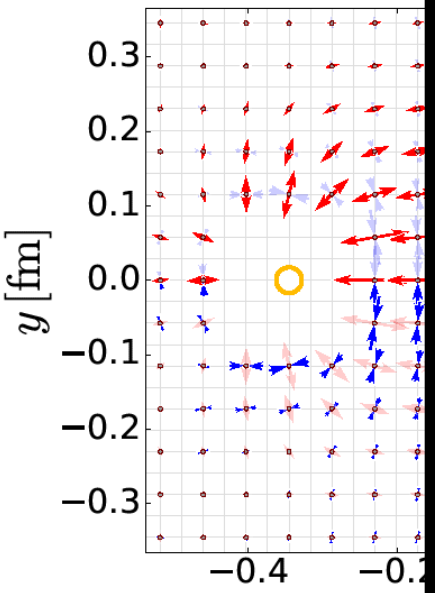

$$F_{\text{stress}} \coloneqq \int_{\text{mid}} \langle T_{zz} \rangle_{Q\bar{Q}} \, d^2x$$

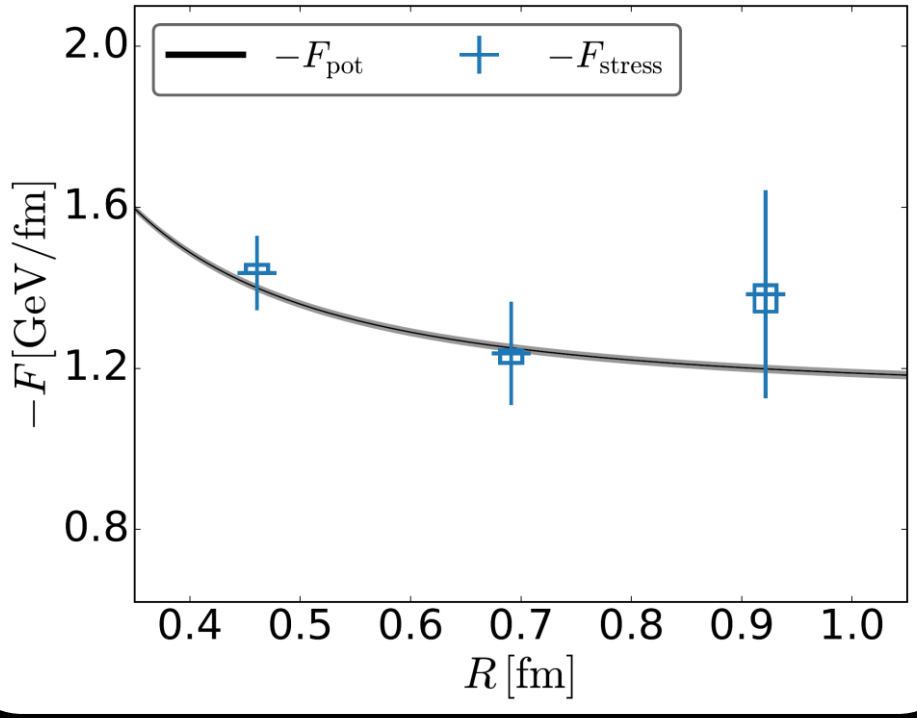
$$V(R) = a + bR + c/R$$

$$F_{\text{pot}} \coloneqq -\frac{dV(R)}{dR}$$

# EMT and confinement potential

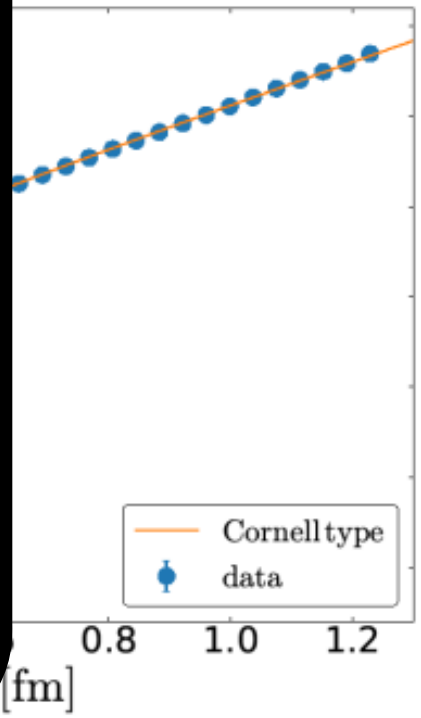
From EMT




Good agreement !!

ntial



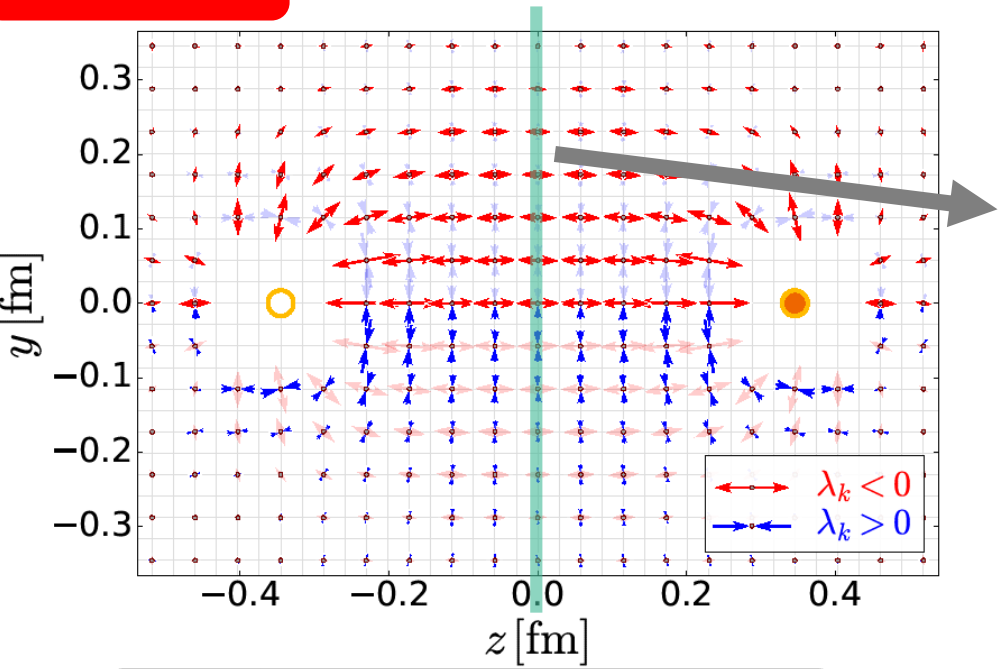

$$F_{\rm stress} \coloneqq \int_{\rm mid} \langle T_{zz} \rangle_{Q\bar{Q}} \, d^2x$$

$$V(R) = a + bR + c/R$$

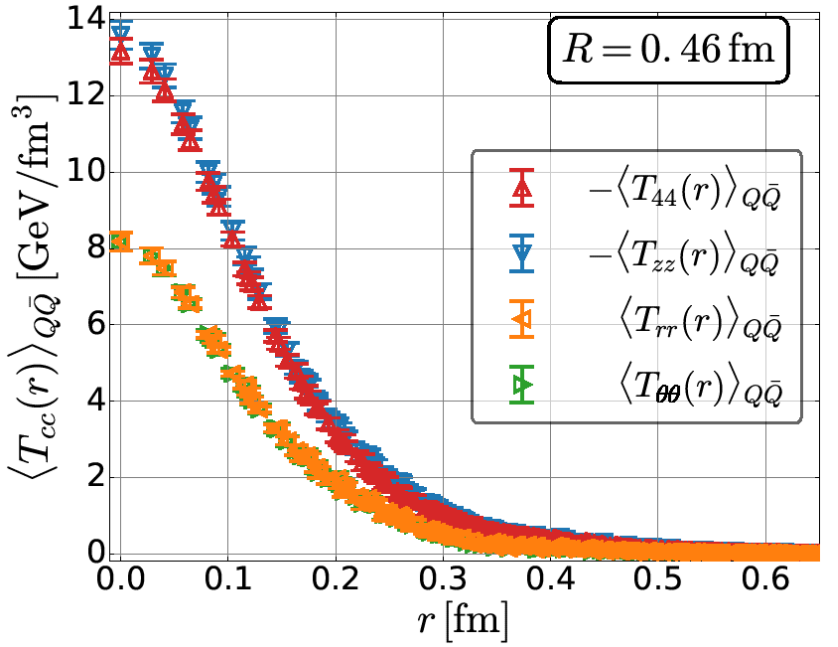
$$F_{\rm pot} \coloneqq -\frac{dV(R)}{dR}$$

# Summary and Outlook

**Summary** First measurement of stress distribution on the lattice!!




Stress distribution b/w $Q\bar{Q}$




Distribution of EMT on mid-plane

**Outlook**

- ✓ We need to explain the stress distribution using, for example, Abelian-Higgs model (RY+ in progress)
- ✓ Application : two flux tubes, finite temperature, full QCD…

# Back up

# Ground saturation

